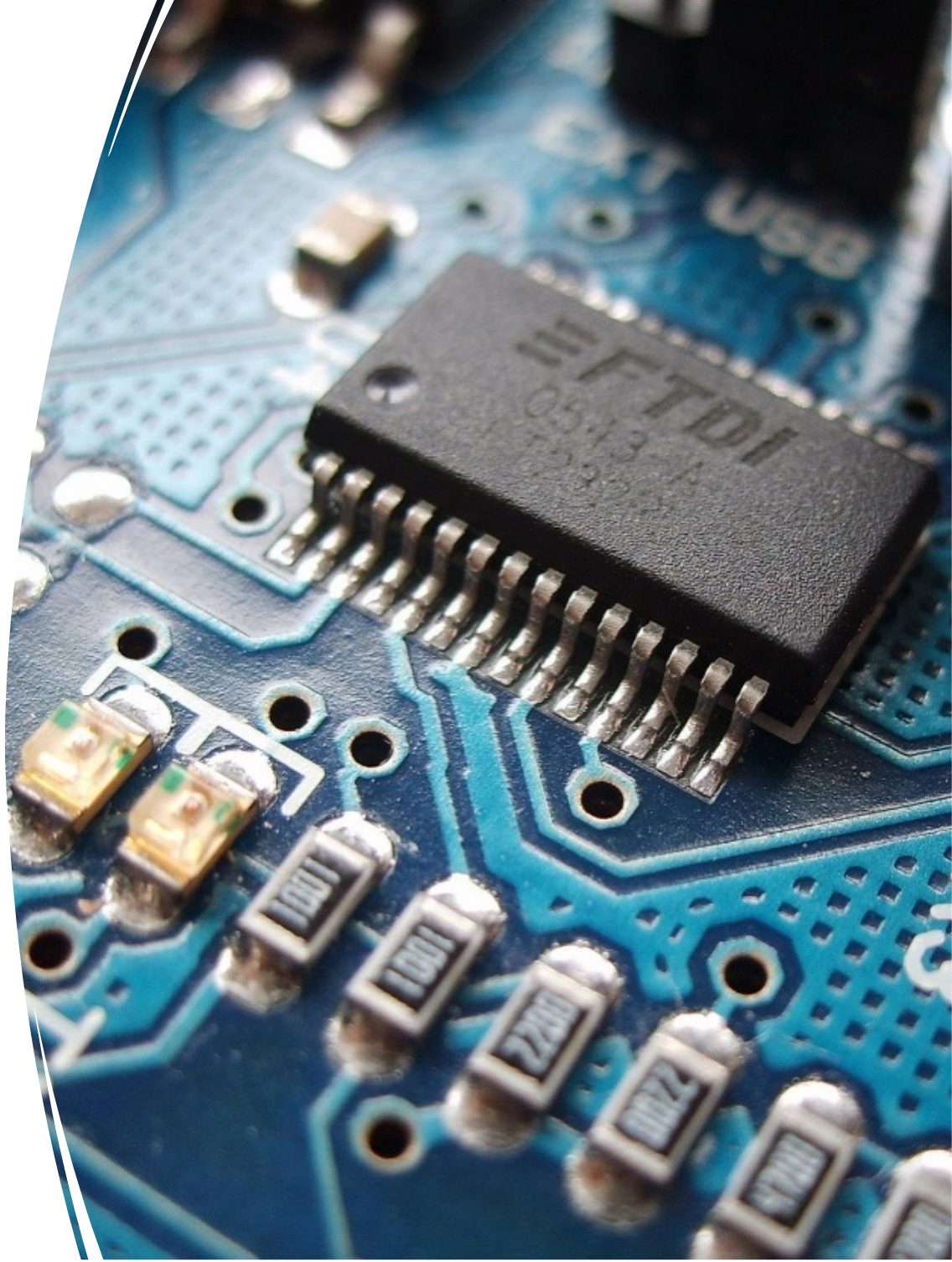
EE 2032 Electronics

- Week 1 - Introduction



Preliminaries

Class schedule:	Monday 15:00 – 17:00 (RTEM2.Z11)
Instructor:	Dr. Kamil Erdayandı, kamil.erdayandi@marmara.edu.tr Office: M2-232, Phone: (216) 777 3535
Teaching Assistant:	Rabia Eda Patoğlu, eda.patoglu@marmara.edu.tr

Syllabus

Recommended Textbook: - R. Boylestad and L. Nashelsky, Electronic Devices and Circuit Theory, 10th Ed., Prentice Hall, 2009.

Attendance: Classroom attendance (%70) is mandatory.

Quizzes: There will be two scheduled quizzes.

Examinations: Exams will be conducted in a closed-book format.

Labs: **Six** labs will be conducted.

- The first two lab will be at the **5th week!**
- Others will be announced later on.
- It is a **must** to attend the labs! Otherwise, you may **fail** the course!
- TA: Rabia Eda Patoğlu, eda.patoglu@marmara.edu.tr

Syllabus

Grading:

Quizzes	10%
Labs	20%
Midterm	30%
Comprehensive Final	40%

Syllabus

Week	Topic	Chapter
1	Introduction	
2	Basic Circuit Principles, Node Analysis, Mesh Analysis	
3	Superposition, Thevenin and Norton Theorems	
4	Semiconductors	
5	Semiconductor Diodes	1
6	Diode Applications	2.1 – 2.6
7	Diode Applications	2.7 – 2.12
8	Midterm Exam	
9	Bipolar Junction Transistor	3
10	DC Biasing BJTs	4.1 – 4.5
11	DC Biasing BJTs	4.6 – 4.11
12	BJT AC Analysis	5
13	Field Effect Transistors	6
14	FET Biasing	7
15	FET Amplifiers	8
16	Final Exam	

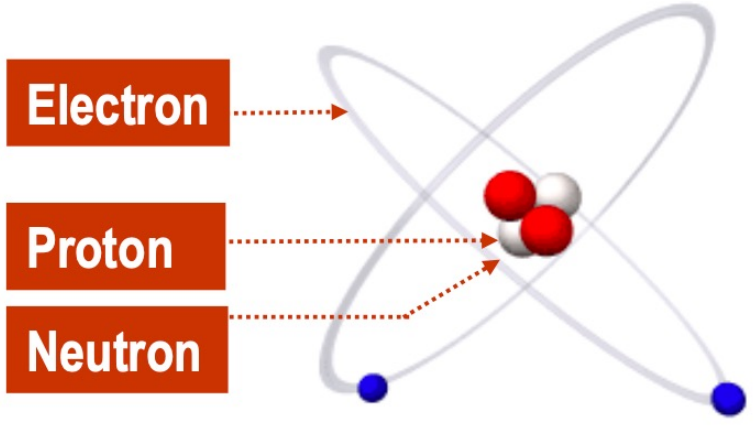
Basic Principles

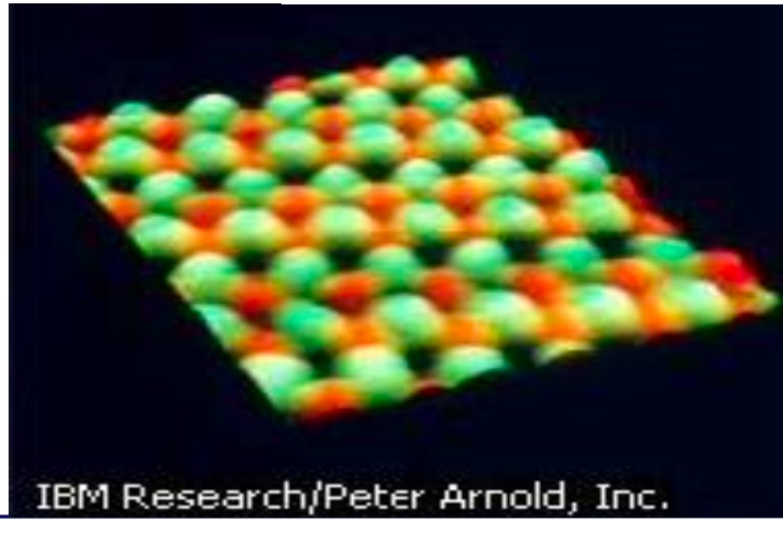
Atom

Structure of atom

Electron is assumed to be negatively charged
Proton is assumed to be positive charged

Helium Atom





Basic Principles

Electrical Charge

Definition

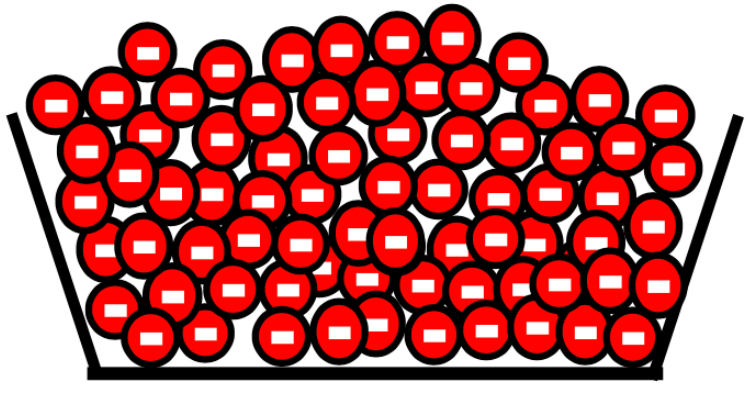
Unit of Electrical Charge Coulomb

6.3×10^{18} electrons $\overset{\Delta}{=}$ 1 Coulomb

or

Electrical charge / electron = $1/(6.3 \times 10^{18})$ Coulomb

$= 1.602 \times 10^{-19}$ Coulomb



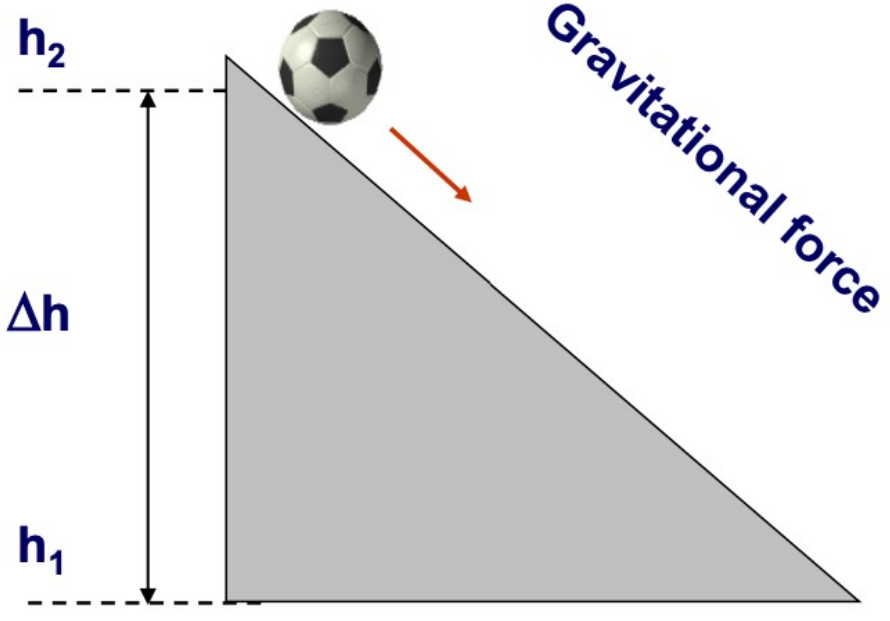
Basic Principles

Basic Principle of Circuit

Mechanical Example

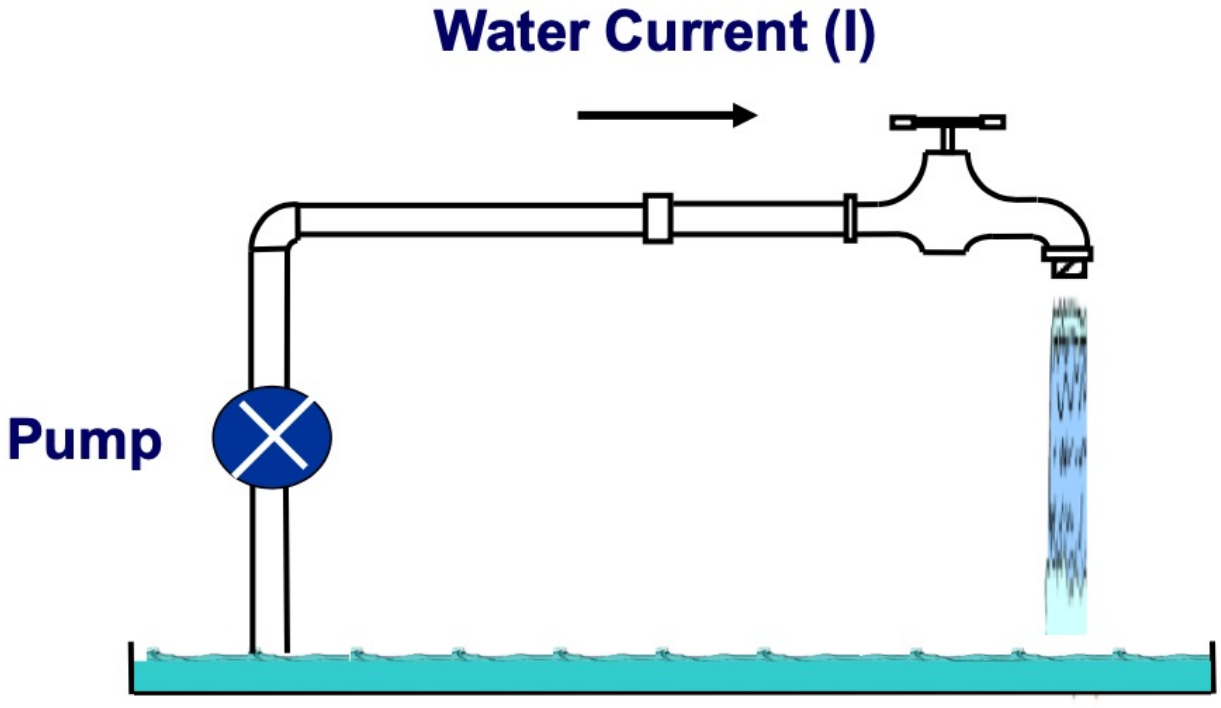


Inclined Surface



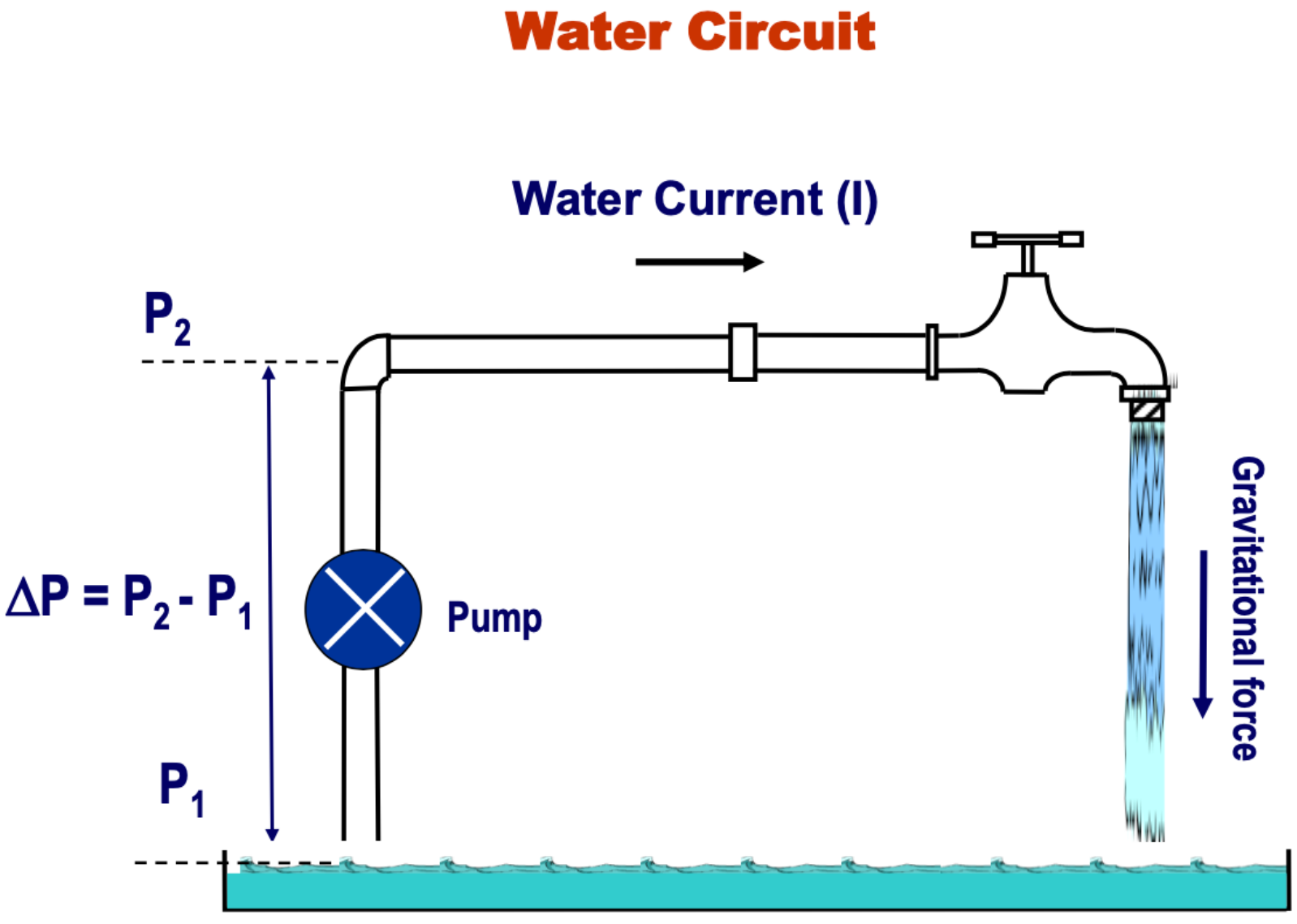
Basic Principles

Water Circuit

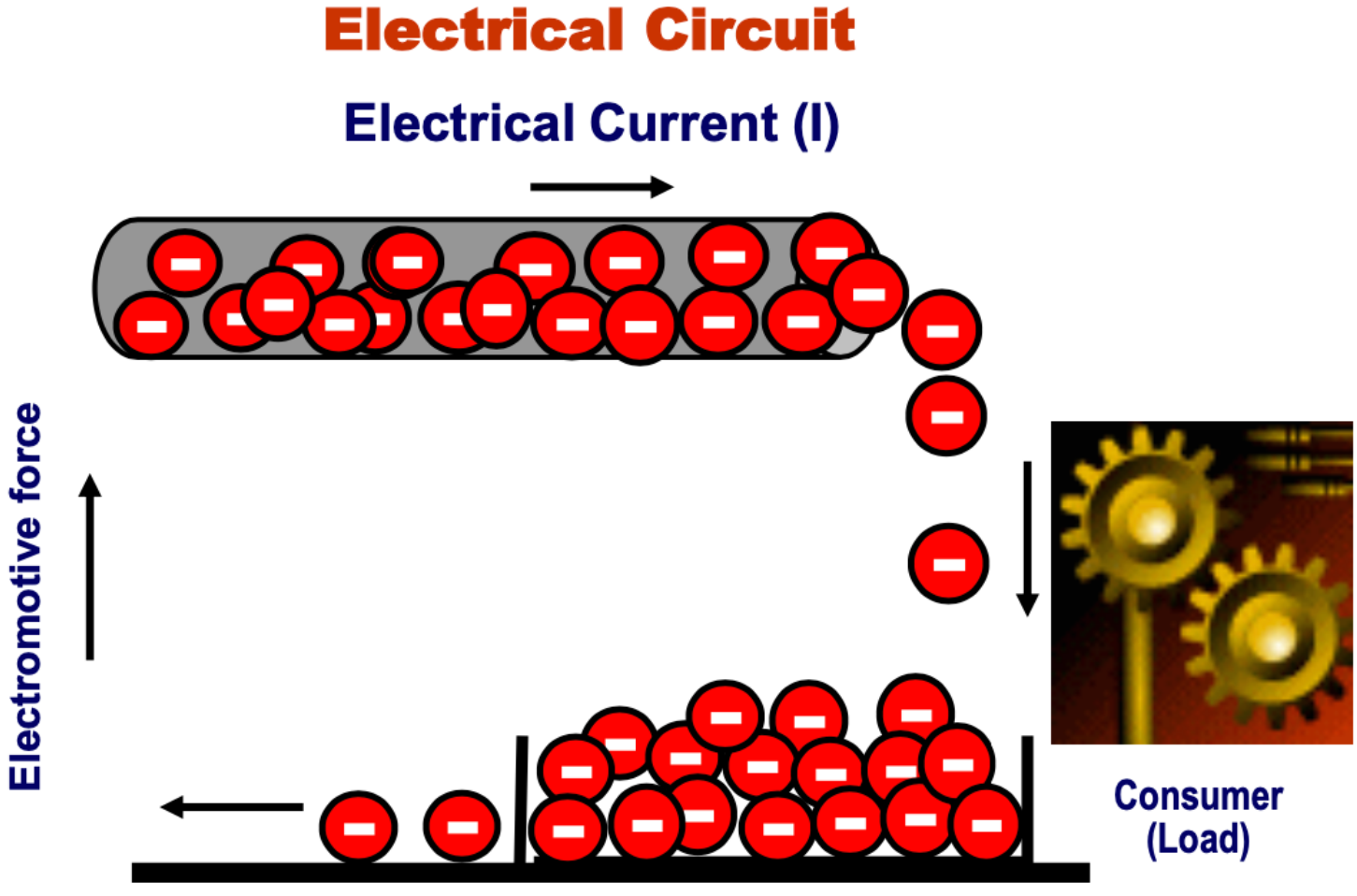


Water Current = Volume (m^3) / sec

Basic Principles

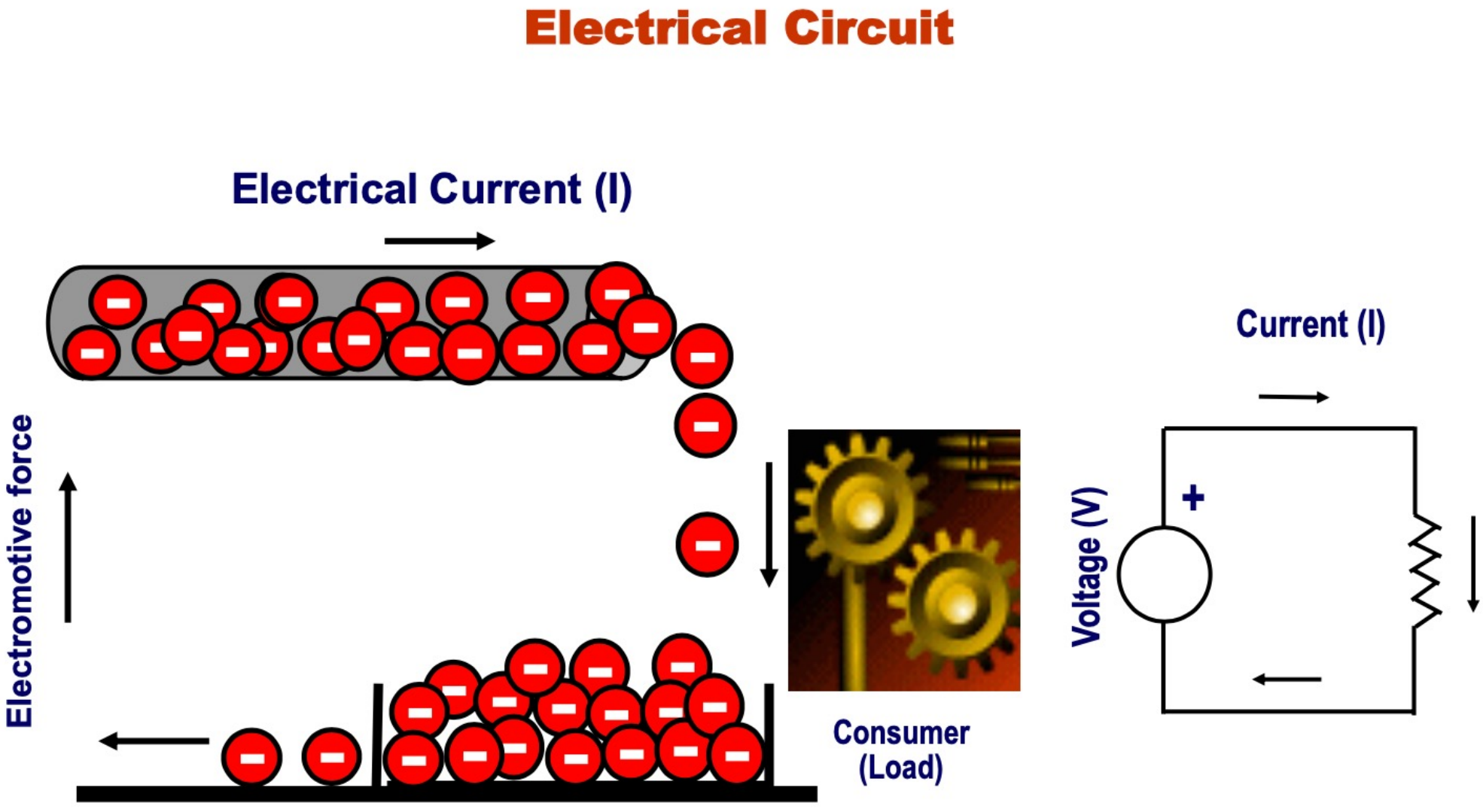


Basic Principles



Electrical Current = No. of electrons / sec
= 1 Coulomb / sec
6.3×10^{18} electrons / sec = 1 Amper

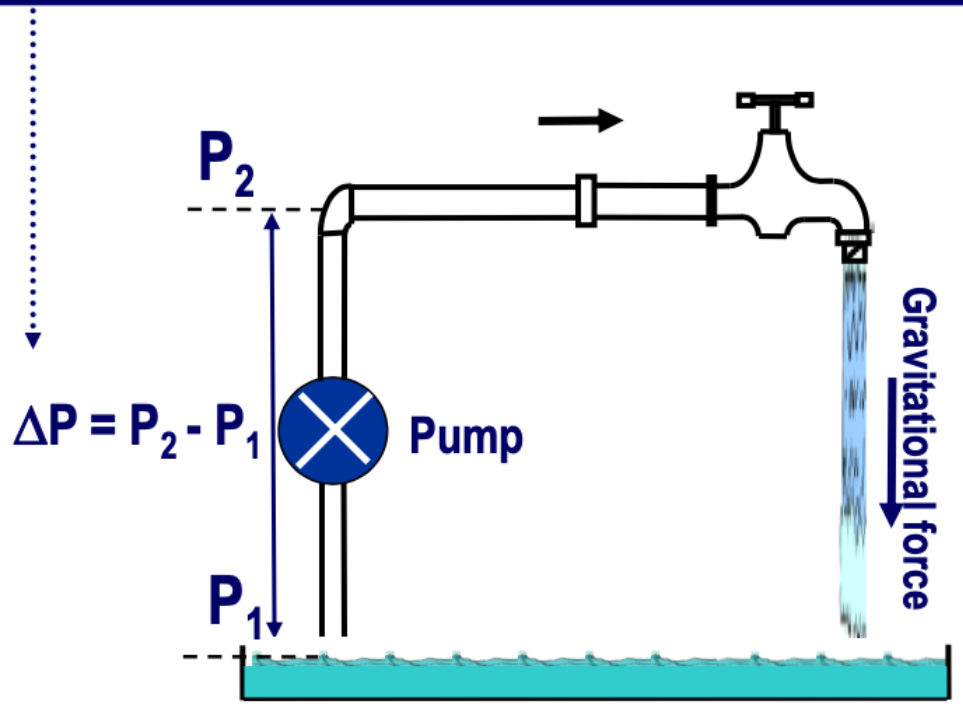
Basic Principles

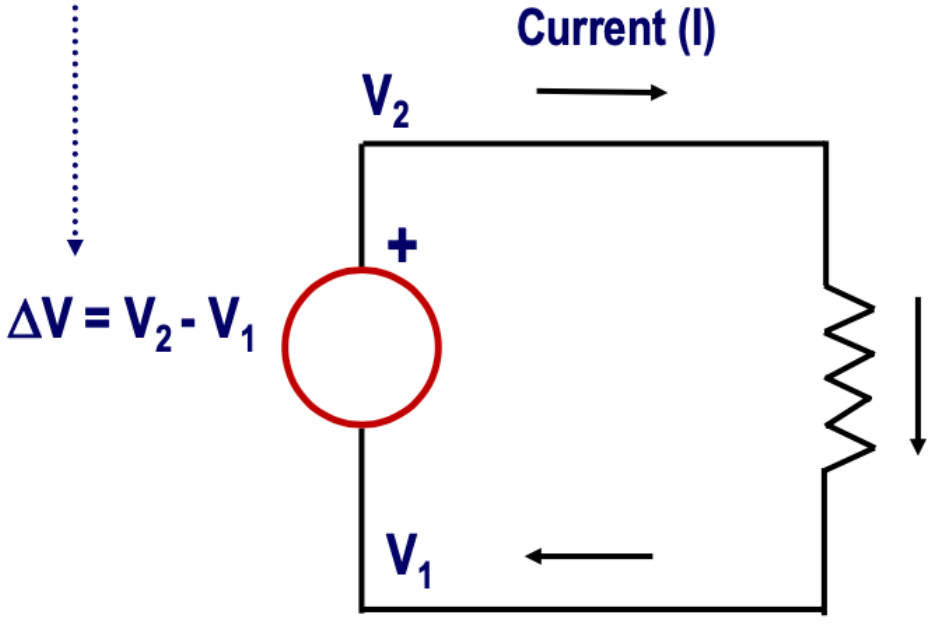


Basic Principles

Voltage Difference

Pressure Difference	Voltage Difference





Basic Principles

Ground Node (Earth Point)

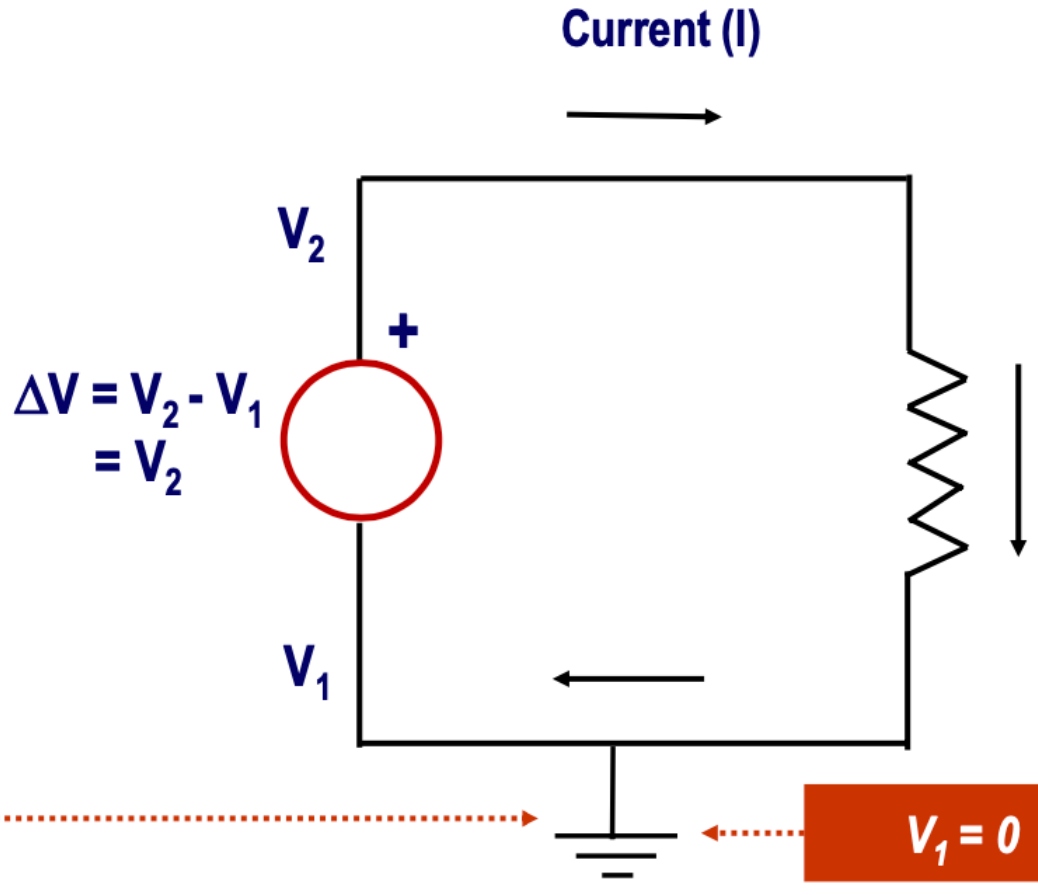
Definition

Ground Node is the point (junction) at which the voltage is assumed to be zero

All other voltages takes their references with respect to this ground node

Representation	*Ground Node*





Basic Principles

Ground Node (Earth Point)

Definition

Ground Node is the point (junction) at which the voltage is assumed to be zero

All other voltages takes their references with respect to this ground node



Ground Node (Black Terminal)

Measured Node (Red Terminal)



Basic Principles

Electrical Current

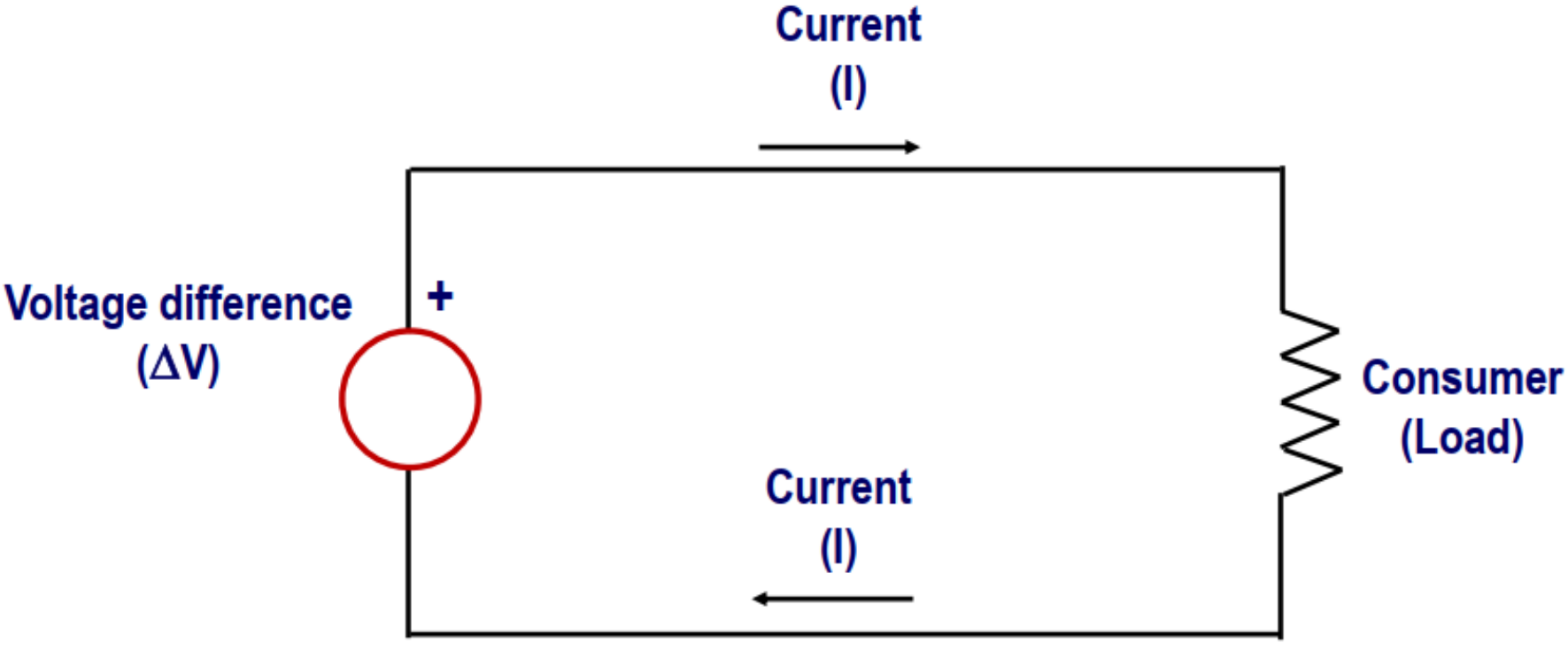
Current = no. of electrons transferred / time duration

$I = \Delta Q / \Delta t$

1 Amp = 1 Coulomb / 1 Seconds

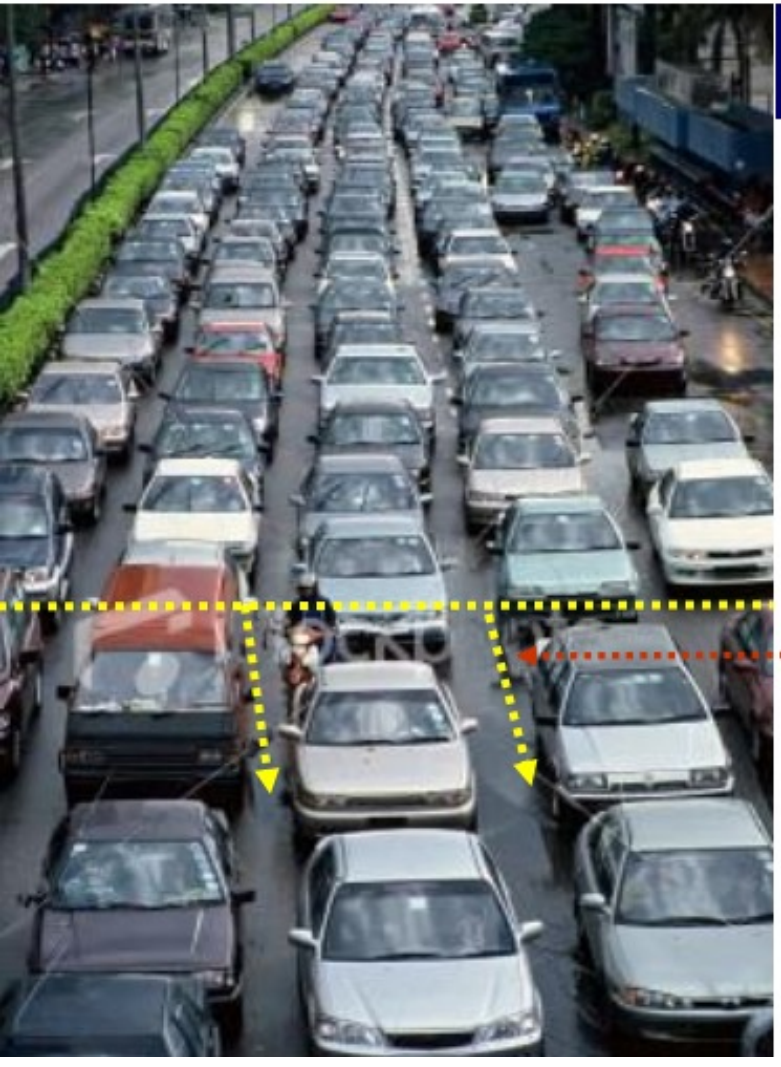
Charge = Current x Time duration

$\Delta Q = I \times \Delta t$



Basic Principles

Traffic Current

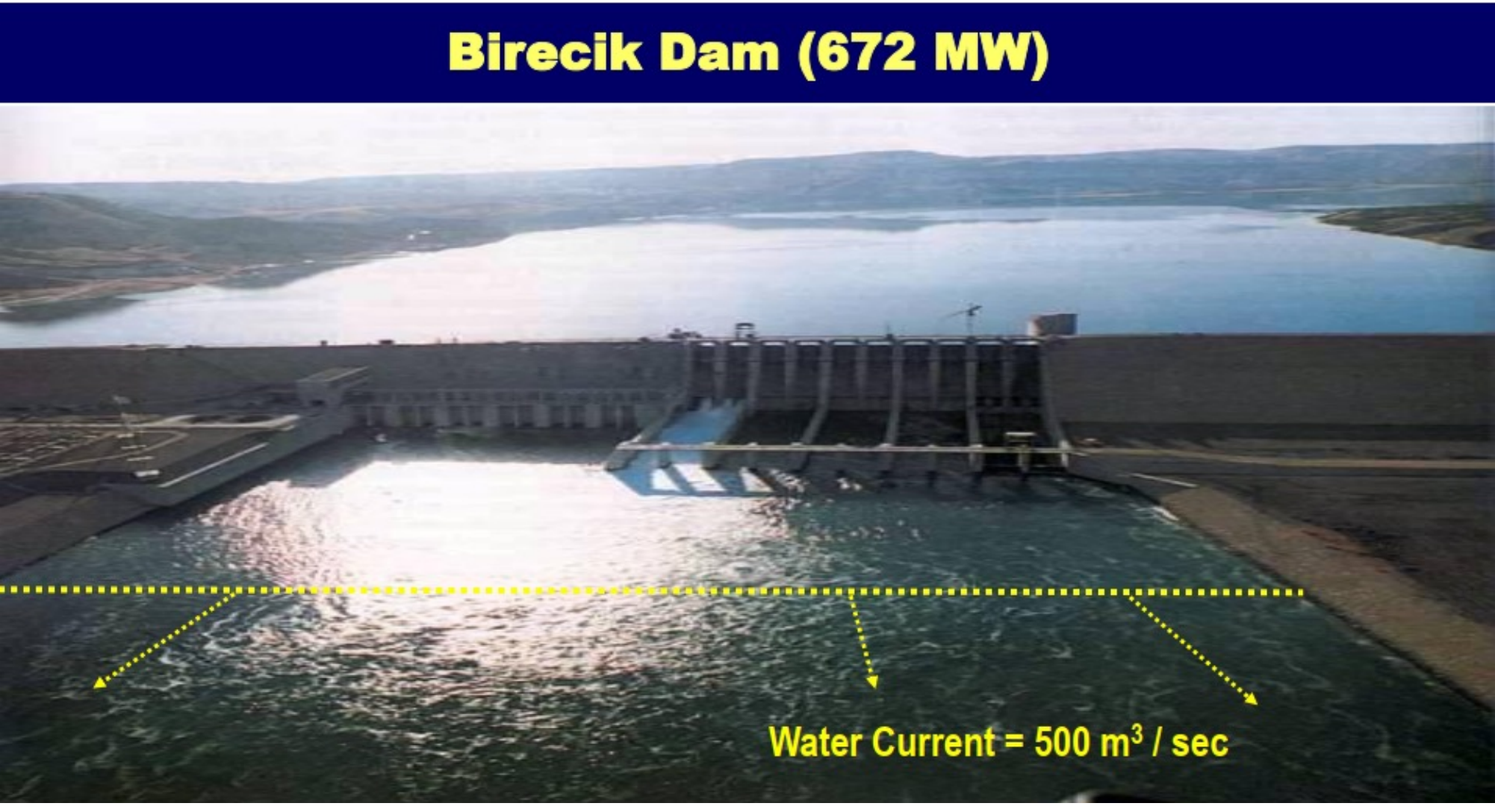


Cars Flowing in a Highway

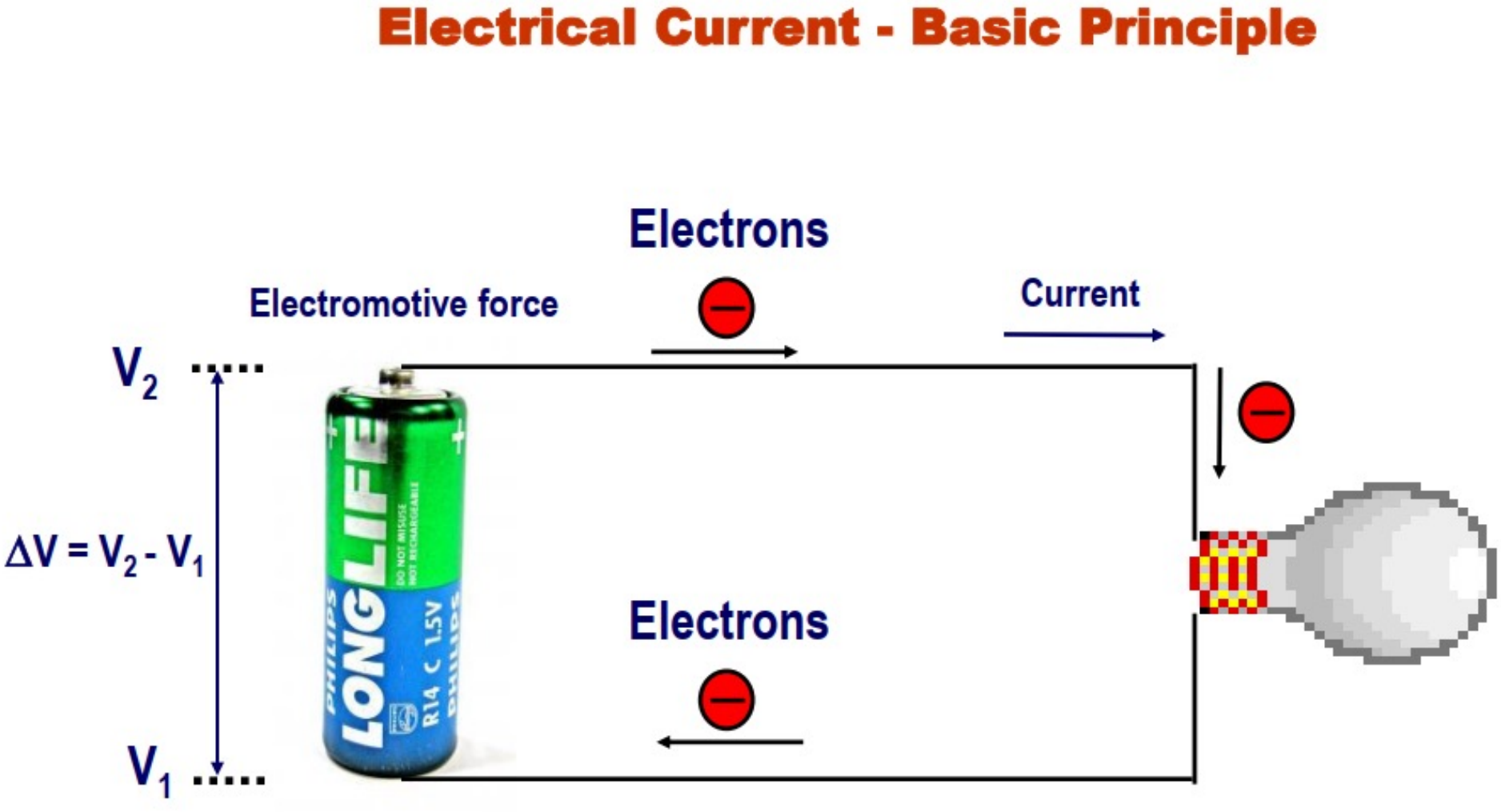
Traffic Current = Cars / minute

Basic Principles

Water Current

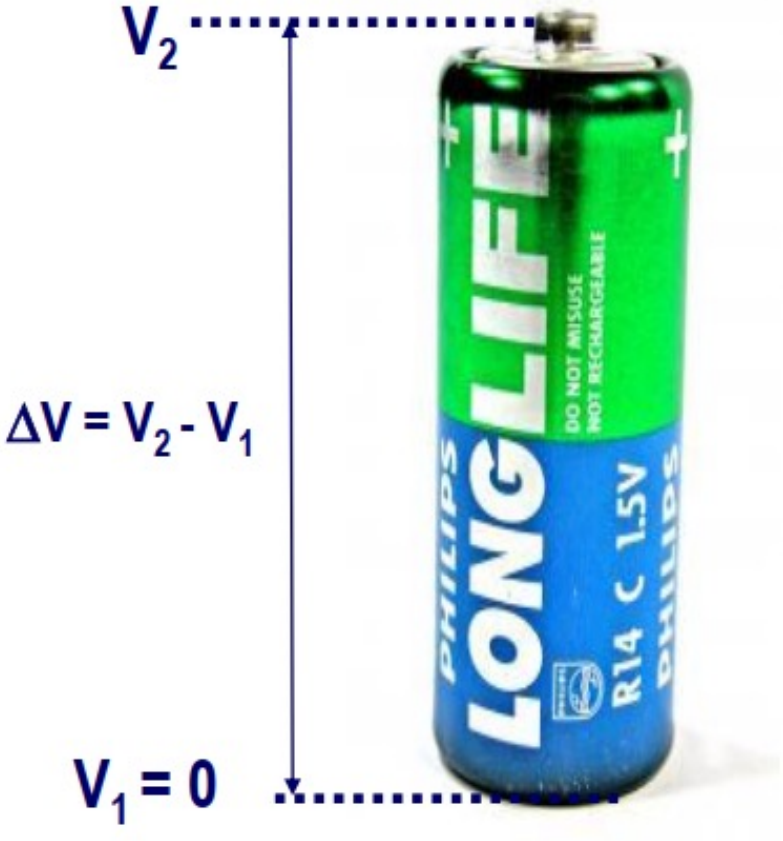


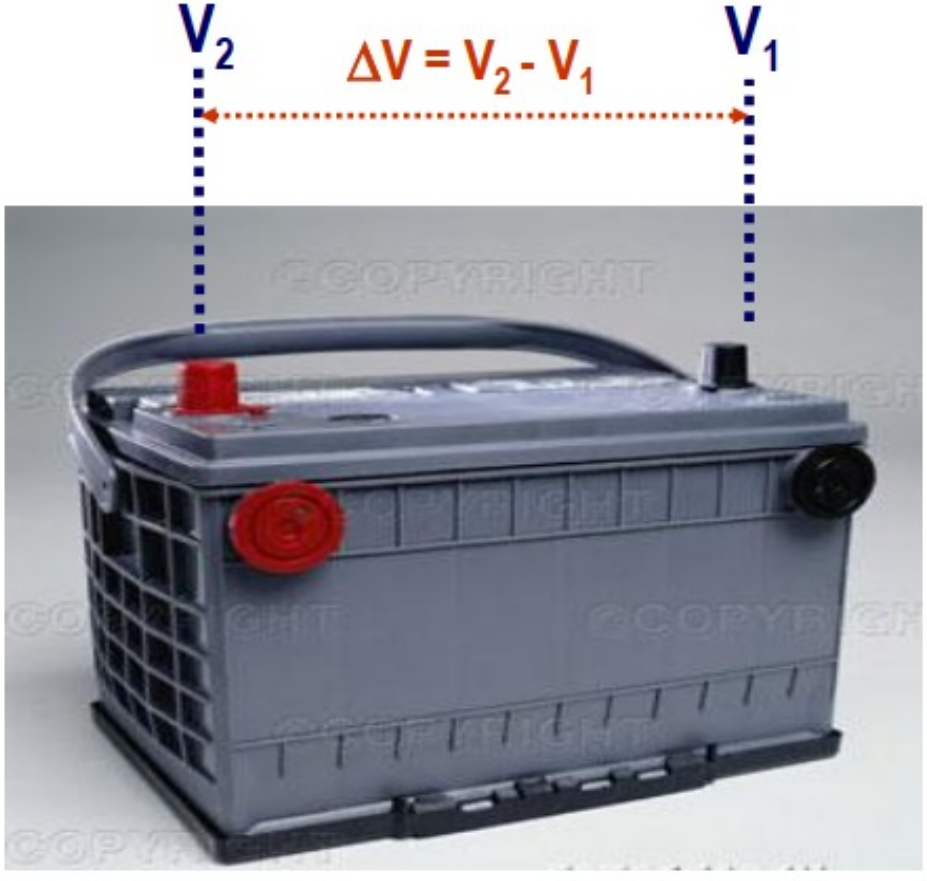
Basic Principles



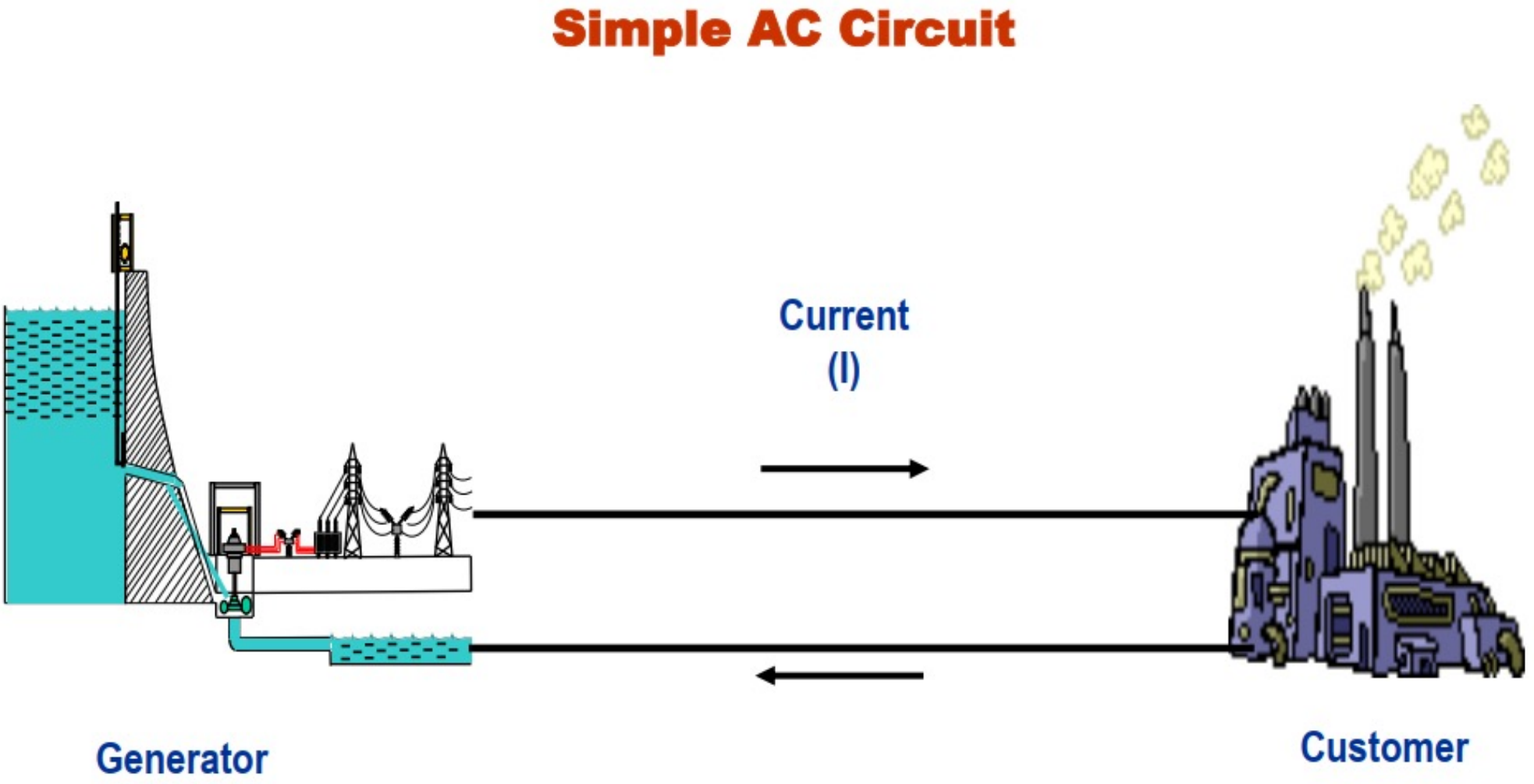
Basic Principles

Electrical Current
DC (Direct Current) Sources





Basic Principles

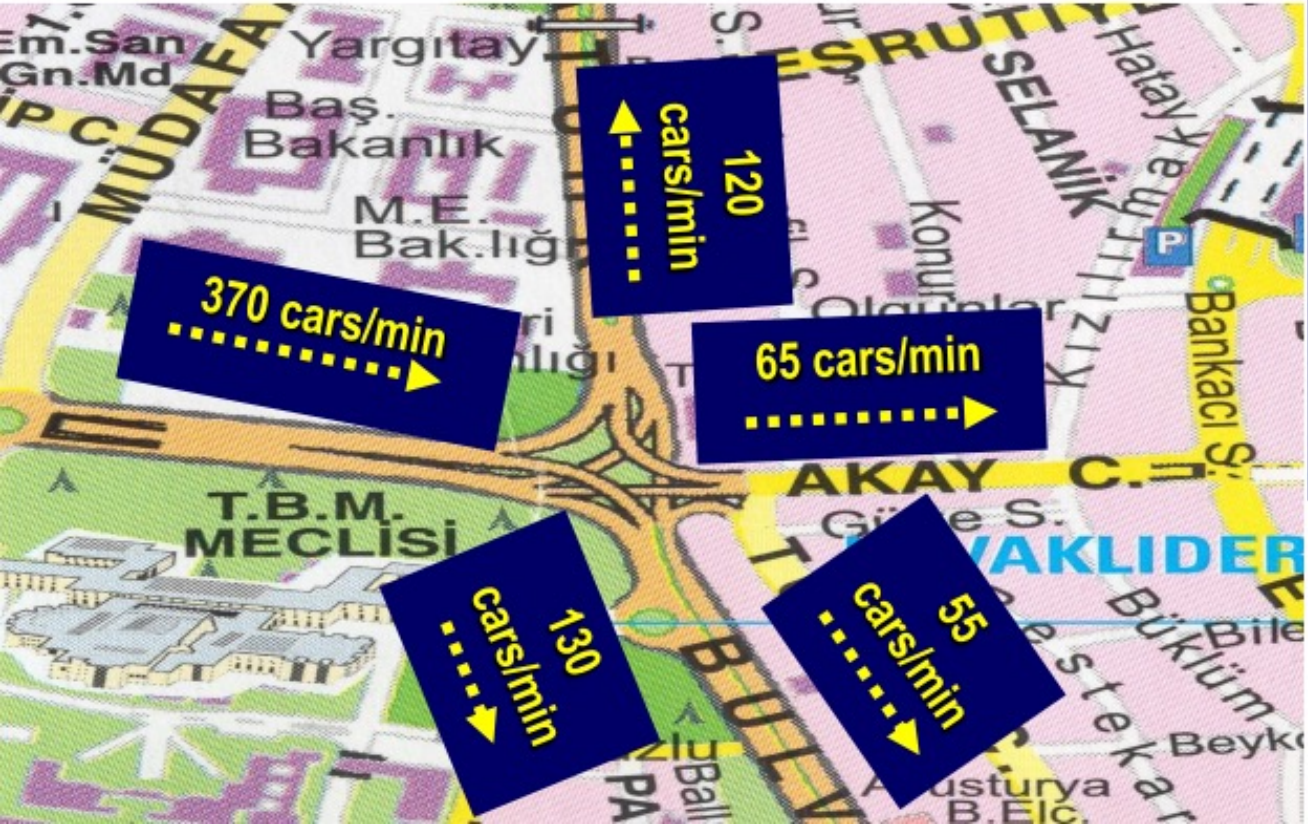


Basic Principles

Kirchoff's Current Law (KCL)

Basic Principle

Σ Cars entering = Σ Cars leaving



Balance

Cars entering: 370

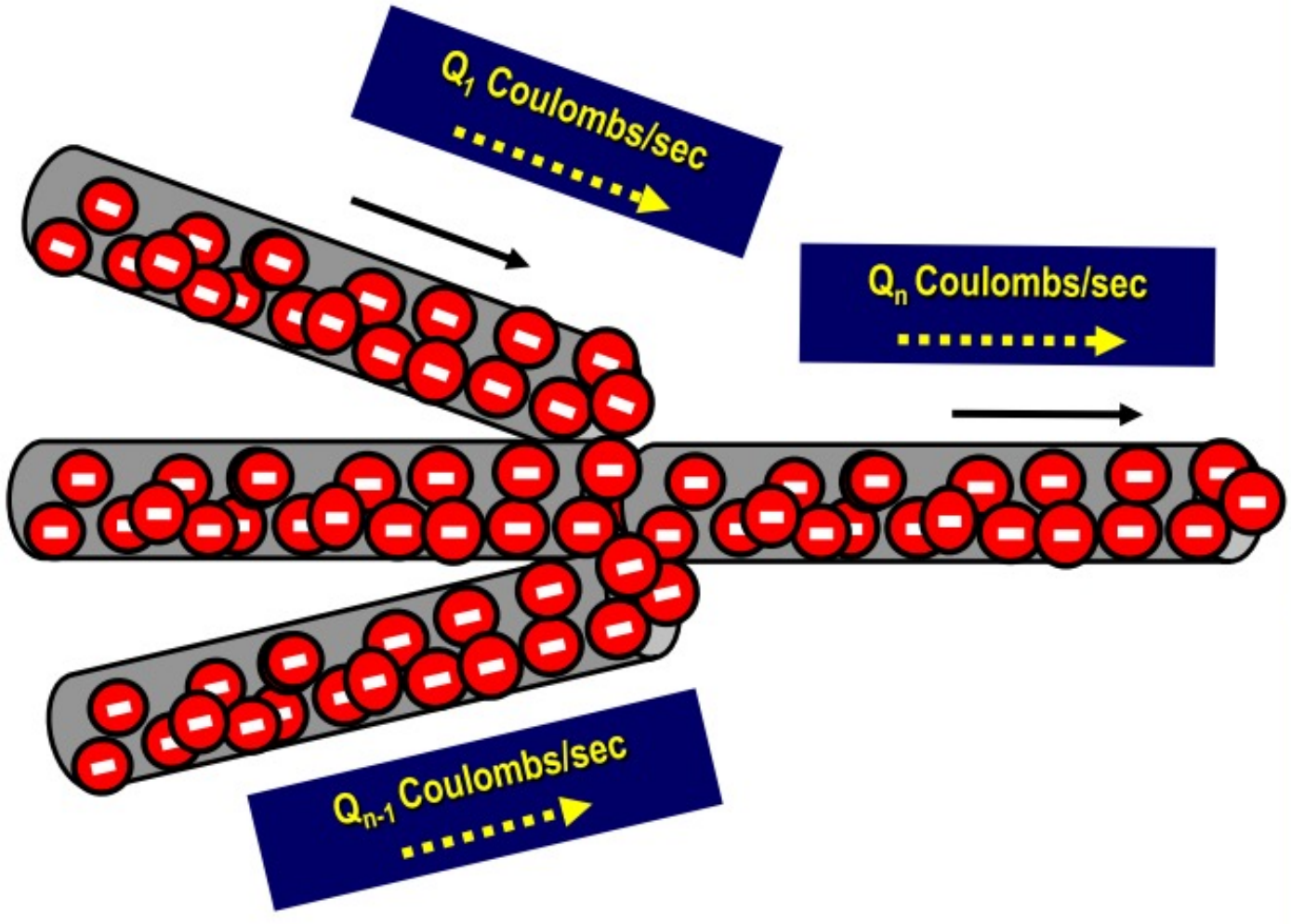
Cars leaving:

120
65
55
130
+

370

Basic Principles

Kirchoff's Current Law (KCL)



Charges entering

Q_1

Q_2

Q_{n-1}

\+

Q_{in}

Charges leaving

$Q_{out} = Q_n$

Balance

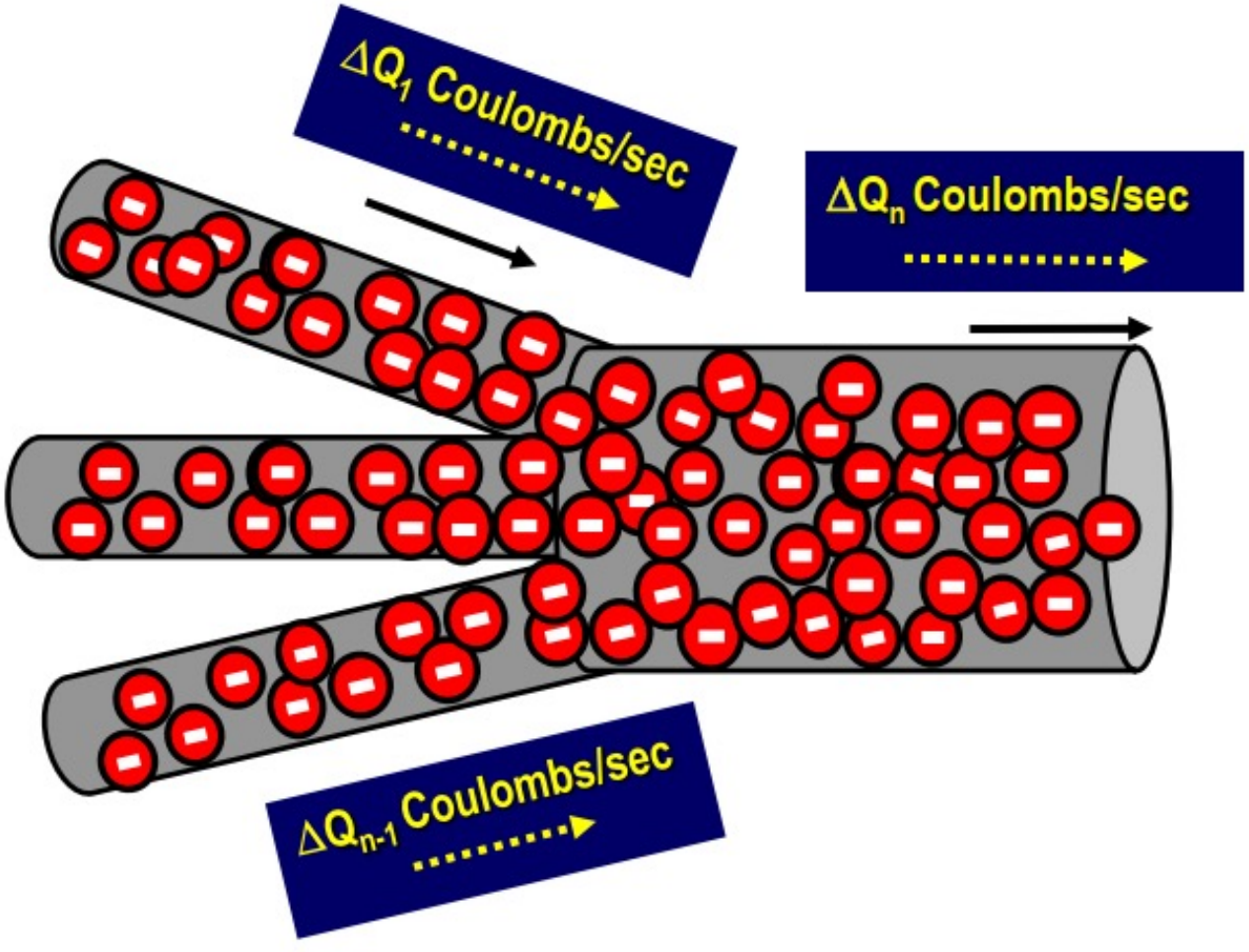
$Q_{in} = Q_{out}$ or

$Q_{in} - Q_{out} = 0$ or

$\Sigma Q = 0$

Basic Principles

Kirchoff's Current Law (KCL)



or

$$\Sigma \Delta Q = 0$$

or

$$\Sigma \Delta Q / \Delta t = 0$$

or

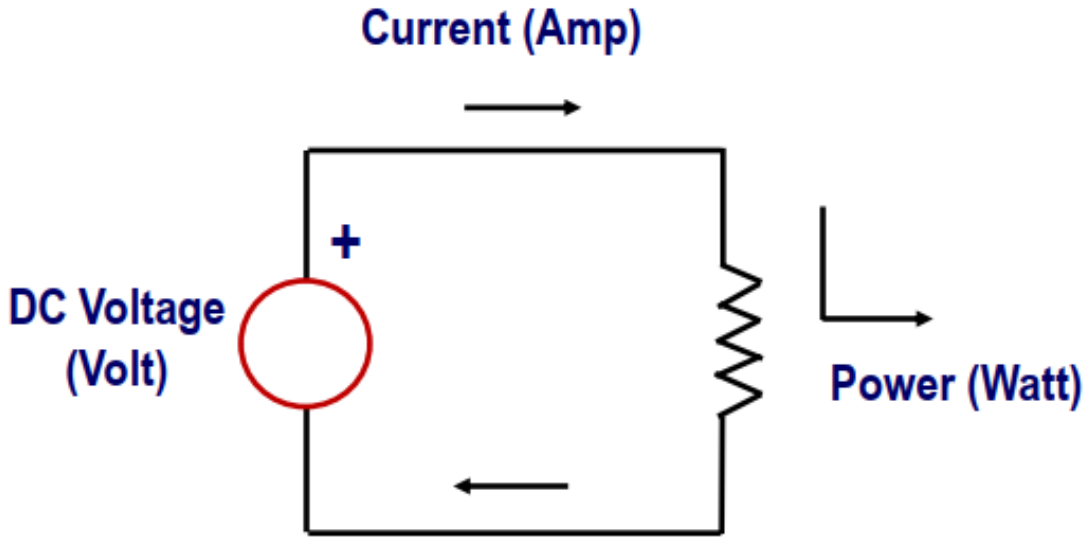
$$\sum_{i=1}^{i=n} I_i = 0$$

Kirchoff's First Law
or
Kirchoffs Current Law

Basic Principles

Electrical Power

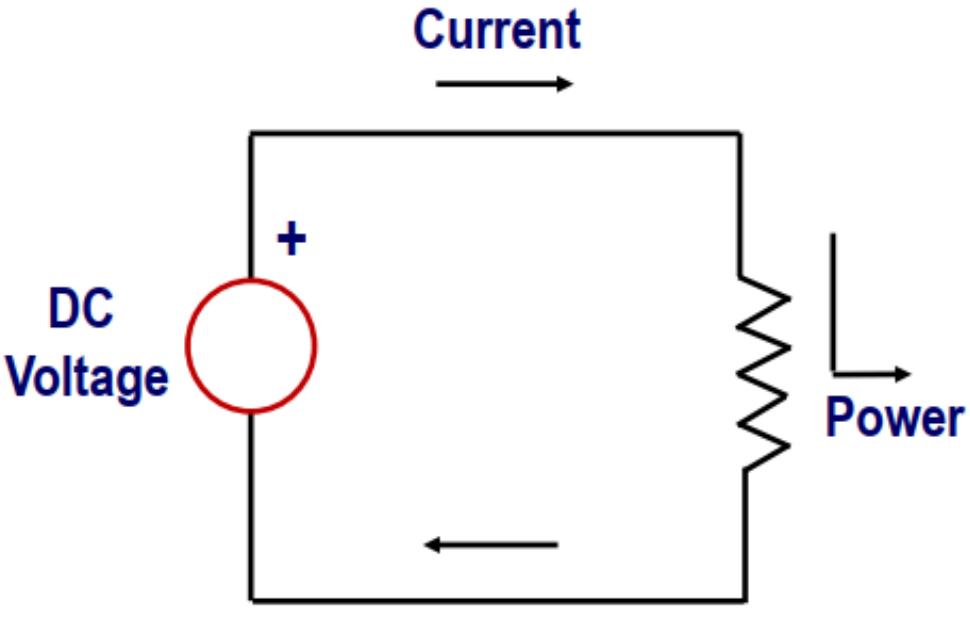
Definition	Power = Voltage x Current
P = V x I (Watt) = (Volt) x (Amp)	



Basic Principles

Voltage

Definition
Power = Voltage x Current *or* *P = V x I*
Voltage = Power / Current *or* *V = P / I*



Basic Principles

Voltage

Definition

Power = Voltage x Current

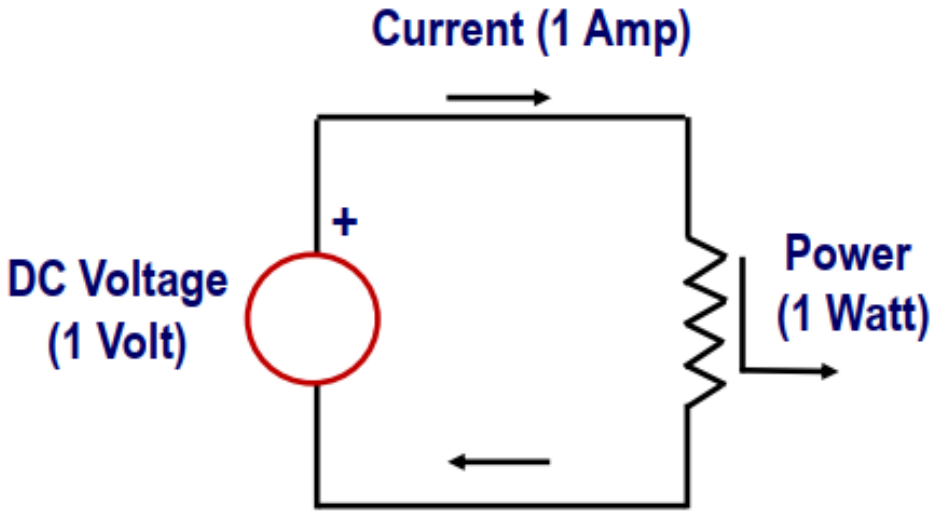
or

Voltage = Power / Current

or

V = P / I

1 Volt = 1 Watt / 1 Amp

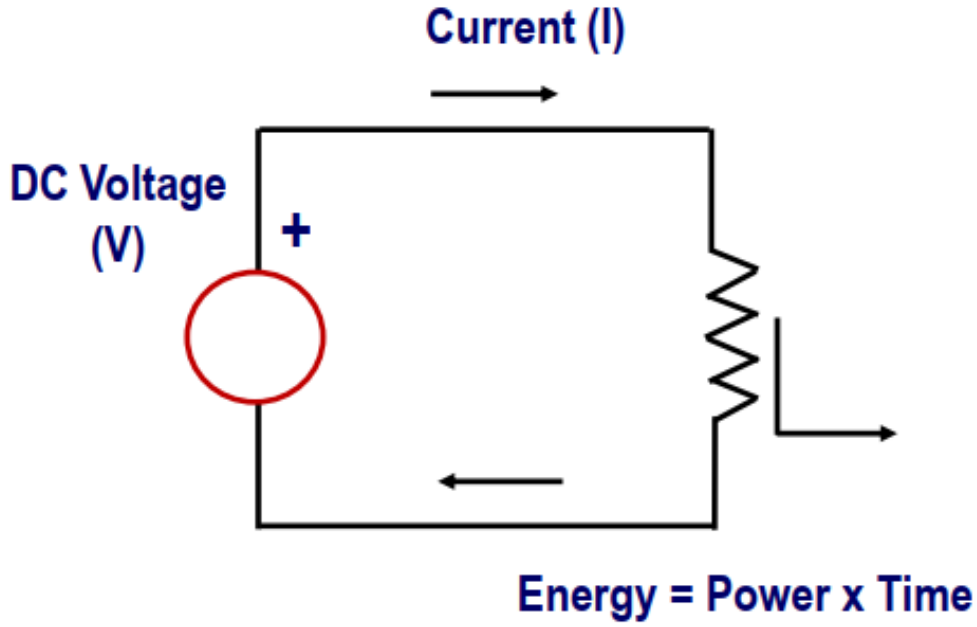


Basic Principles

Electrical Energy

Definition

Energy = Power x Time
(Watt-sec) (Watt) (second)



Basic Principles

Unit of Electrical Energy

Definition

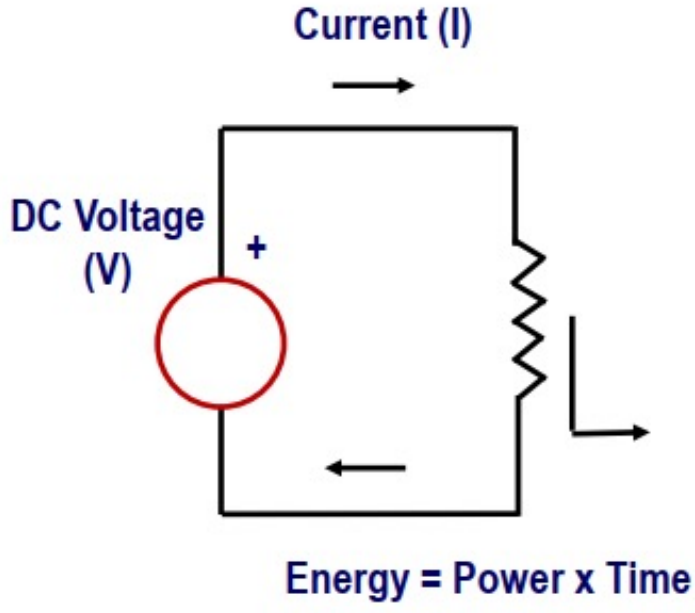
Energy = Power x Time
(Watt-sec) (Watt) (second)

Energy = Power x Time
(KiloWatt-hour) (KiloWatt) (hour)

x 1000 *x 1000* *x 3600*

1 KiloWatt - hour = 1000 x 3600 Watt x seconds
= 3 600 000 Joules

1 KiloWatt = 1000 Watts
1 Hour = 3600 seconds



Basic Principles

Electrical Energy

Example

Calculate the monthly payment for the energy consumed by the lamp shown

Source voltage is 220 Volt
Current drawn by the lamp is 5 Amp
Price of electrical energy is 12 Cents / kWh

Power = Voltage x Current

P = V x I

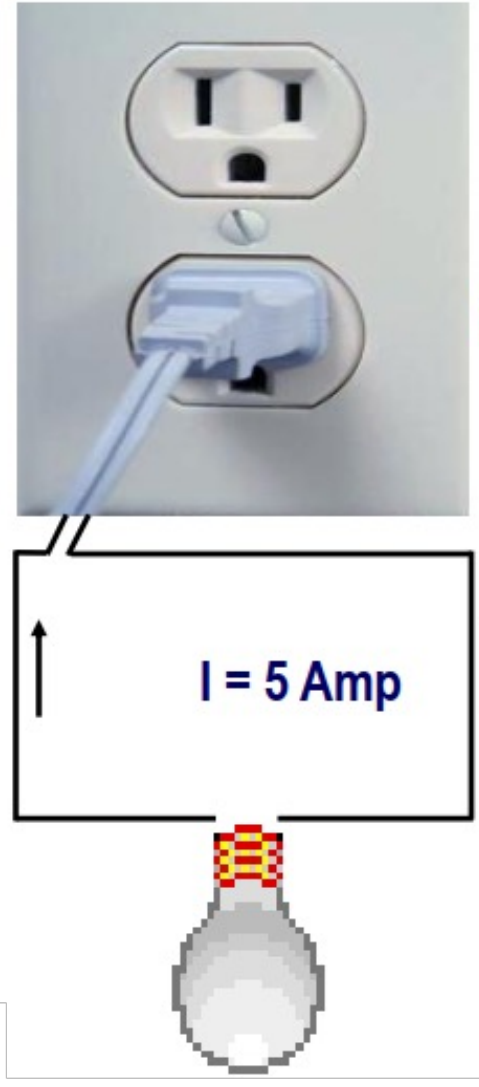
P = 220 x 5 = 1100 Watts

Energy = P x Δt

= 1100 Watts x (24 hours /day x 30 days/month)

= 792000 Watt hours = 790.2 kWh

Monthly payment = 790.2 x 12 Cents / month



Basic Principles

Alternative Definition of Voltage

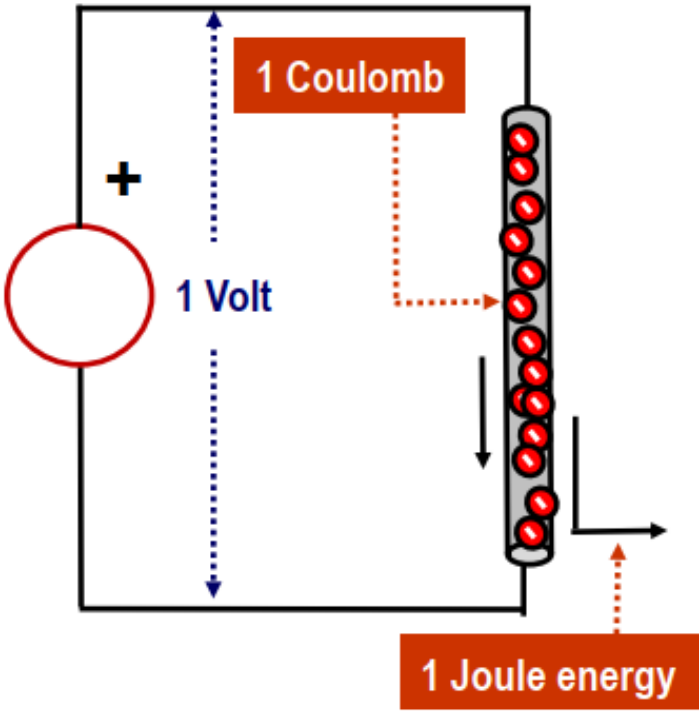
1 Volt = *1 Watt / 1 Amp*
= *(1 Joule /sec) / 1 Amp*
= *1 Joule / (1 Amp x sec)*
= *1 Joule / 1 Coulomb* (*)

(*) *Remember that 1 Amp = 1 Coulomb / 1 sec*

1 Volt is the voltage needed;

- **to move 1 Coulomb of electrical charge,**
- **to spend 1 Joule of energy for this movement**

in a conductor



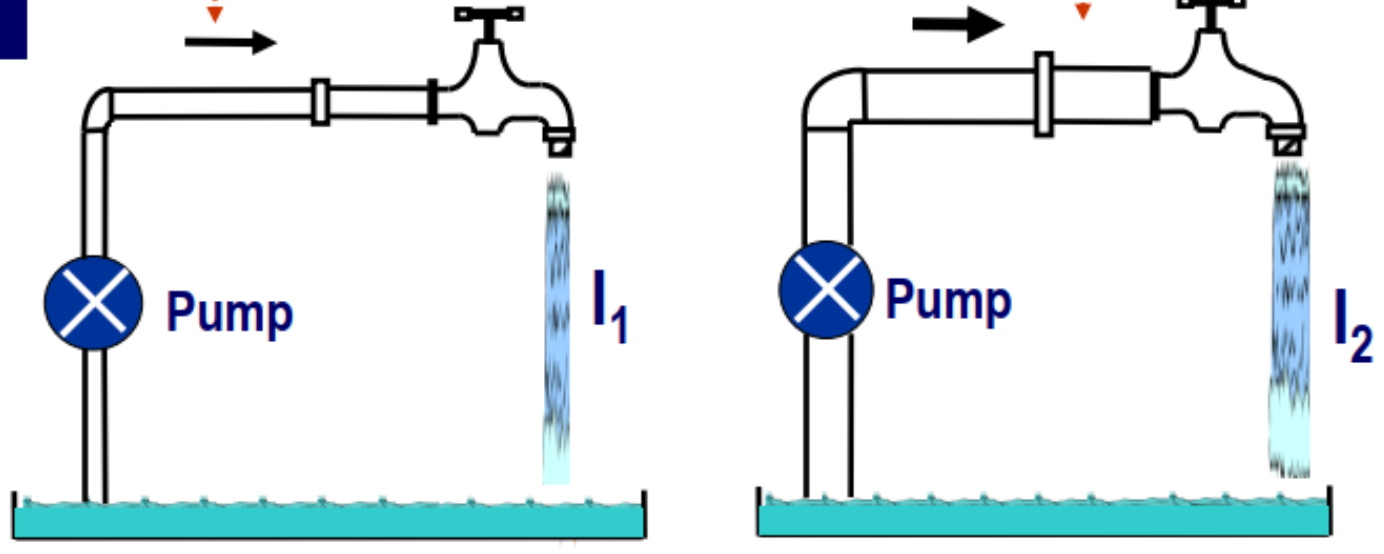
Basic Principles

Resistance

Definition

Resistance is the reaction of a pipe against water flow

Resistance R_1 → $R_1 > R_2$ ← Resistance R_2



Current I_1 → $I_2 > I_1$ ← Current I_2

Basic Principles

Resistance

Definition

Resistance is the reaction of a conductor against electrical current



Resistance R_1
Current I_1



Resistance R_2
Current I_1



$R_1 > R_2$ $I_1 < I_2$

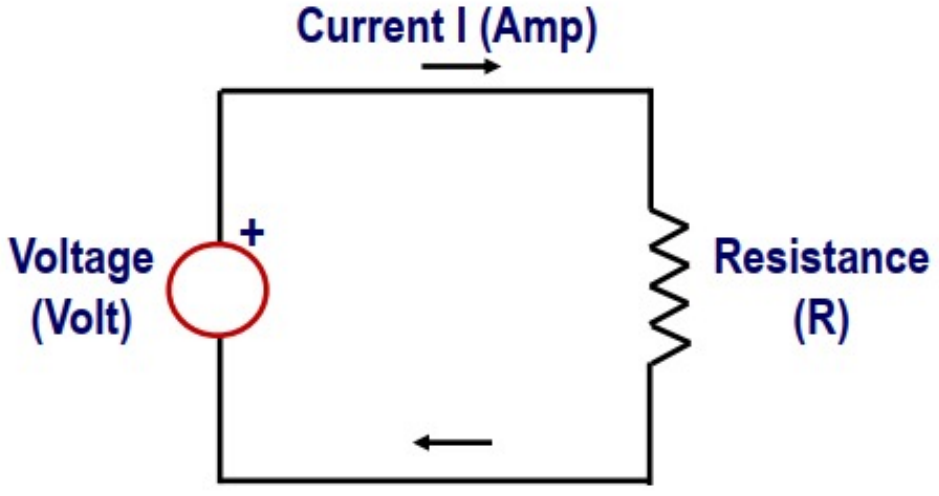
Basic Principles

Ohm Law

Basic Principles

Current flowing in the circuit is;

- proportional to voltage,
- inversely proportional to resistance



Hence

$$I \text{ (Amp)} = V \text{ (Volt)} / R \text{ (Ohm)}$$

or

$$V \text{ (Volt)} = R \text{ (Ohm)} \times I \text{ (Amp)}$$

Unit of resistance is Ohm

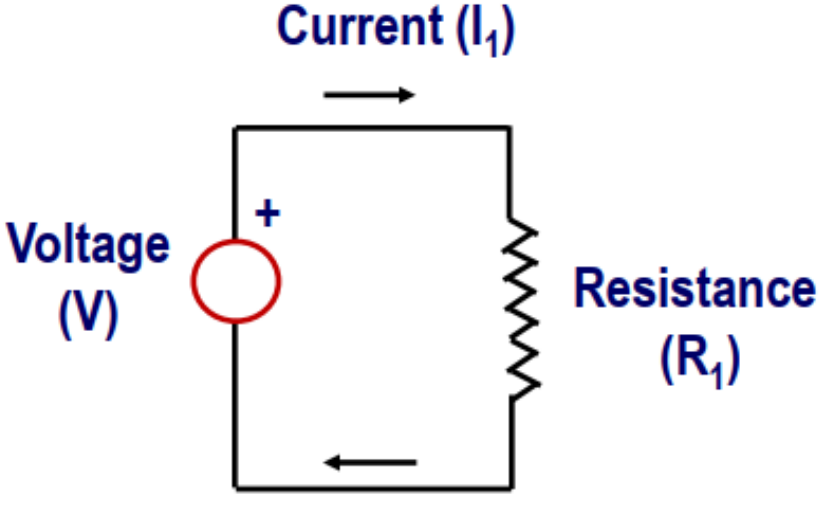
1 Ohm is the resistance that allows 1 Amper to pass at 1 Volts voltage;
1 Ohm = 1 Volt / 1 Amper

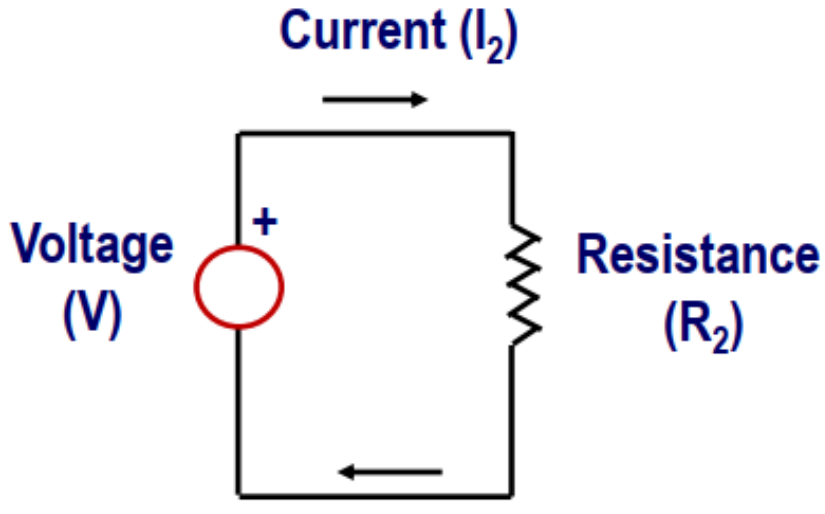
Basic Principles

Ohm Law

Two circuits with different Resistances, identical voltage sources

Resistance R_1	$R_1 > R_2$	Resistance R_2

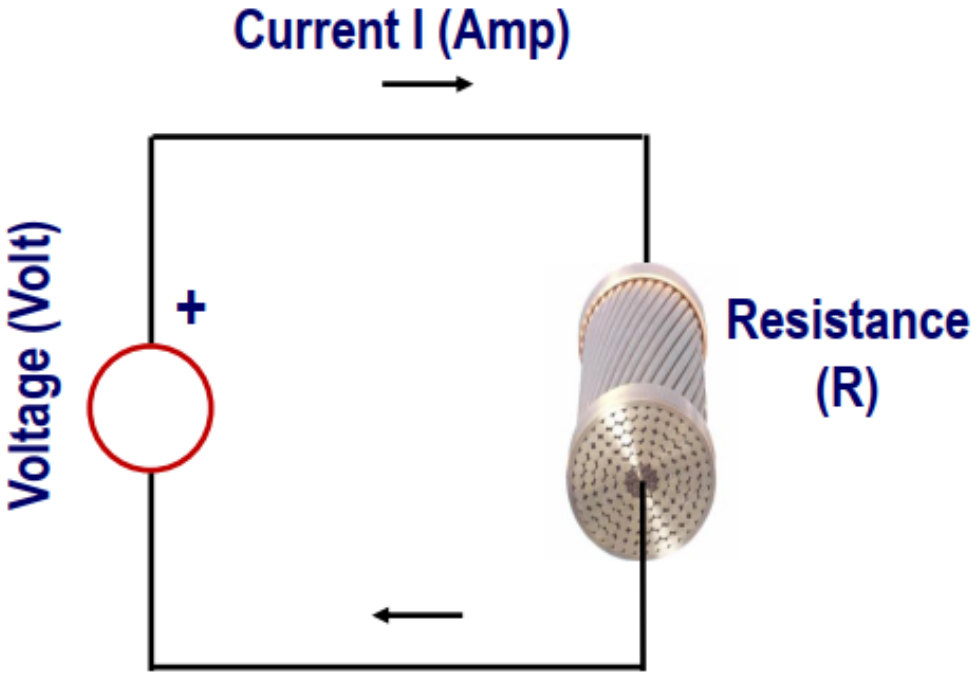




$V / R_1 = I_1$	$I_1 < I_2$	$V / R_2 = I_2$

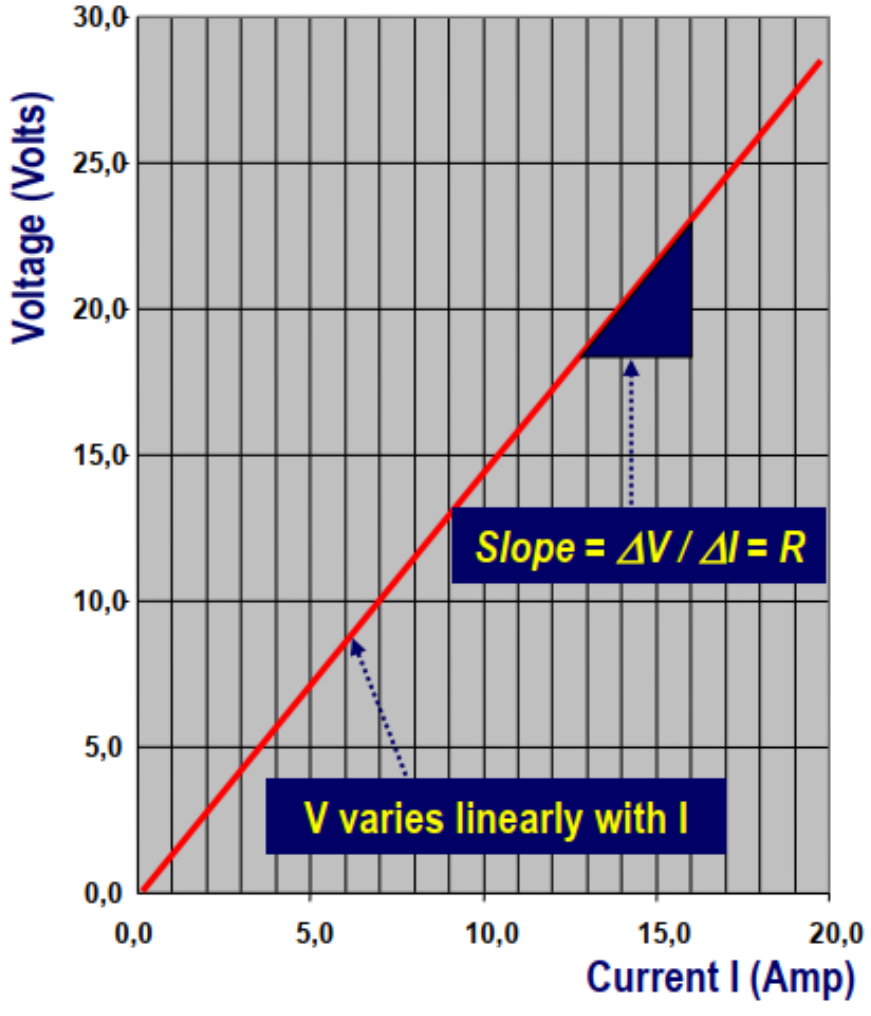
Basic Principles

Ohm Law
V-I Characteristics



$$V = R \times I$$

(Volt) (Ohm) (Amp)



Basic Principles

Ohm Law - Example

Question

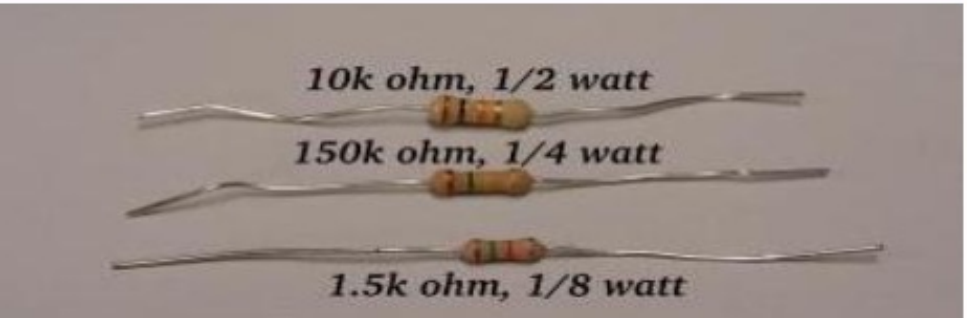
Calculate the current flowing in the circuit shown on the RHS

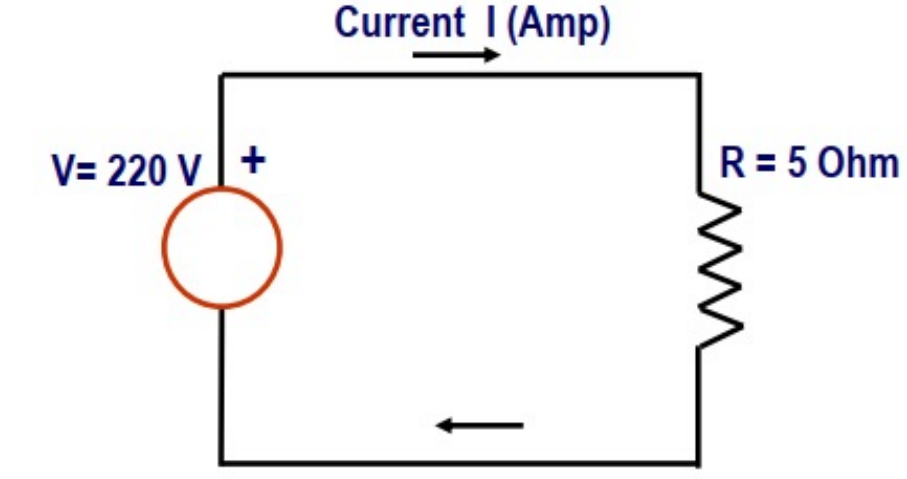
$V = R \times I$

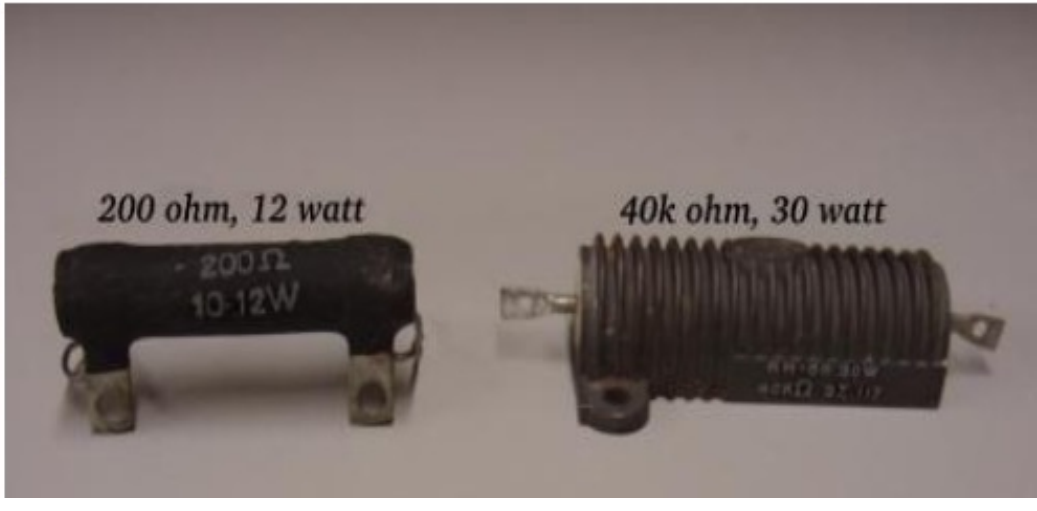
(Volt) = (Ohm) x (Amp)

$I = V / R$

$= 220 / 5 = 44$ Amps







Basic Principles

Resistance Formula

Resistance Formula

$$R = \rho \ell / A$$

where, R is the resistance of conductor,

ρ is the resistivity coefficient,

ρ = 1 / 56 Ohm-mm^2/m (Copper)

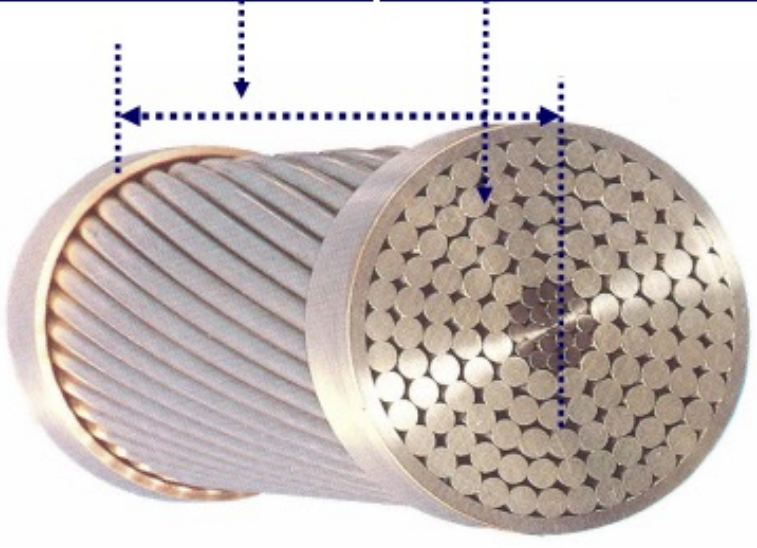
1 / 32 Ohm-mm^2/m (Aluminum)

ℓ (m) is the length of the conductor

A (mm^2) is the cross sectional area of the conductor

ACSR Conductor
(Aluminum Conductor Steel Reinforced)

ℓ (meter) | A (mm^2)



Basic Principles

Resistance Formula

Resistance Formula

Resistance of a cable is proportional to the length and inversely proportional to the cross sectional area of the cable

$$R = \rho \ell / A$$

where, R is the resistance of conductor,

ρ is the resistivity coefficient,

ρ = 1 / 56 Ohm-mm^2/m (Copper)

1 / 32 Ohm-mm^2/m (Aluminum)

ℓ (m) is the length of the conductor

A (mm^2) is the cross sectional area of the conductor



Basic Principles

Resistivity Coefficients of Various Metals

Formula

ρ = 1 / 56 Ohms/meter (Copper)
= 0.0178571 Ohm-mm^2/m

$$R = \rho \ell / A$$

where, R is the resistance of conductor,
ρ is the resistivity coefficient,
ρ = 1 / 56 Ohm-mm^2/m (Copper)
1 / 32 Ohm-mm^2/m (Aluminum)
ℓ (m) is the length of the conductor
A (mm^2) is the cross sectional area of the conductor

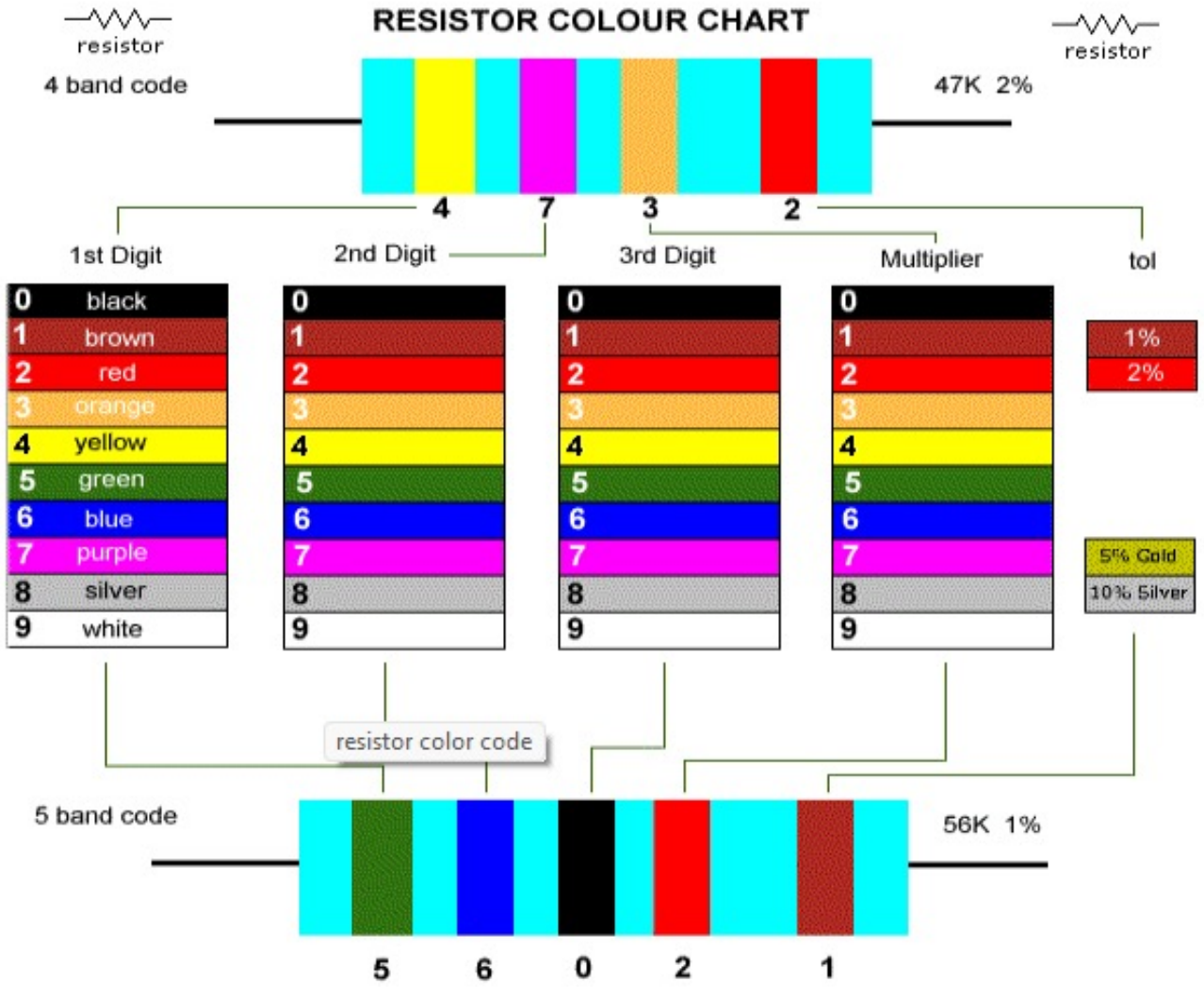
Resistivity Coefficients

Material	Resistivity Coefficient	Resistance
	Ohm-mm^2/m	Ohms/feet
Silver	0.0162	0.00094
Copper	0.0172	0.00099
Gold	0.0244	0.00114
Aluminum	0.0282	0.00164
Mercury	0.9580	
Brass	0.0700	0.00406
Nickel	0.7800	0.00452
Iron	0.1000	0.00579
Platinium	0.1000	0.00579
Steel	0.1180	0.00684
Lead	0.2200	0.01270

Basic Principles

Color Codes for Resistances

Rule



Basic Principles

Power dissipation in a Resistance

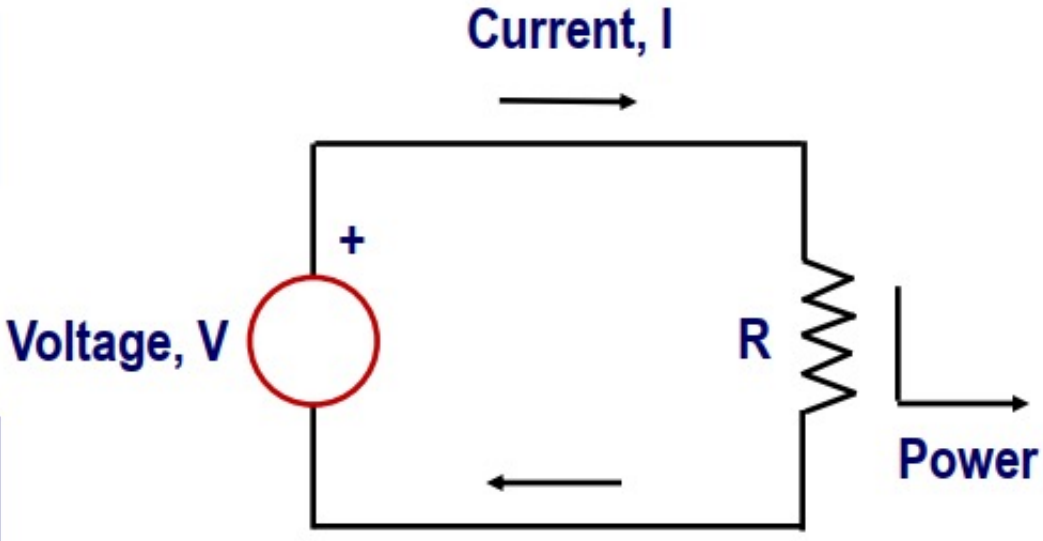
$$V = R \times I$$
$$(Volt) = (Ohm)\ (Amp)$$

On the other hand, it was shown in this lecture that;

$$Power = Voltage \times Current$$
$$or$$
$$P = V \times I$$

Hence, power dissipation in resistance R is

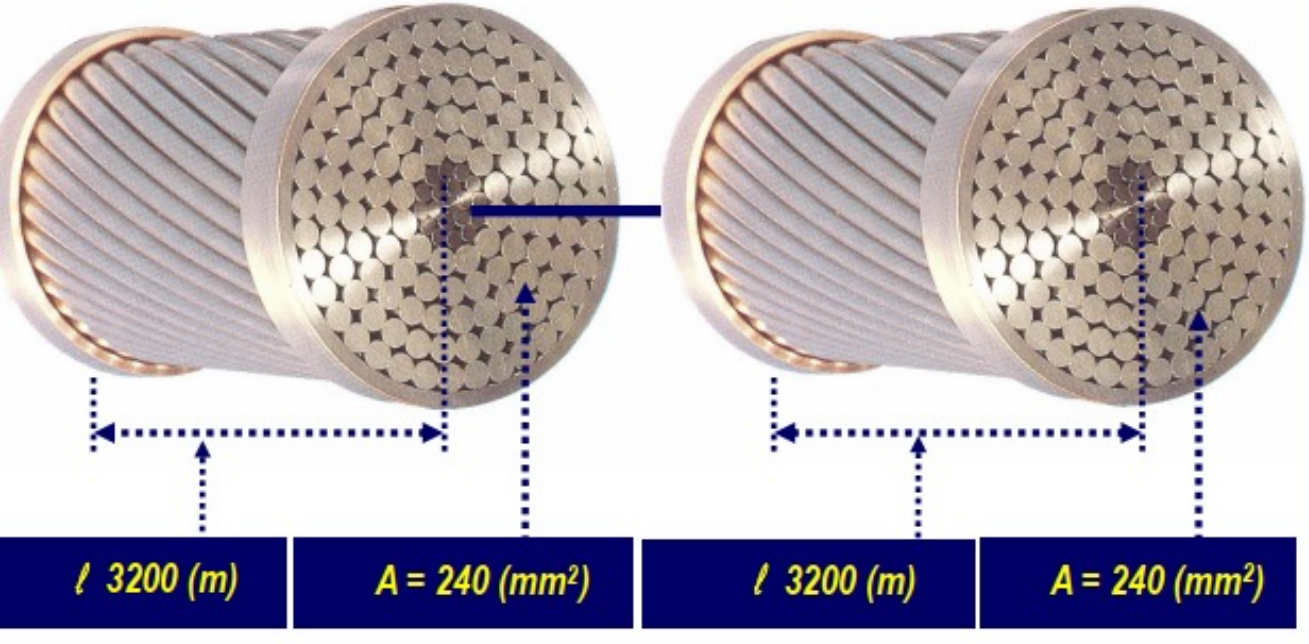
$$Power = R \times I \times I$$
$$= R \times I^2 \quad Watt$$



Basic Principles

Series Connected Resistances

Equivalent Resistance Formula



$R_1 = \rho \ell_1 / A_1$

$R_2 = \rho \ell_2 / A_2$

Let $A_1 = A_2$

Hence;

$\ell_{total} = \ell_1 + \ell_2$

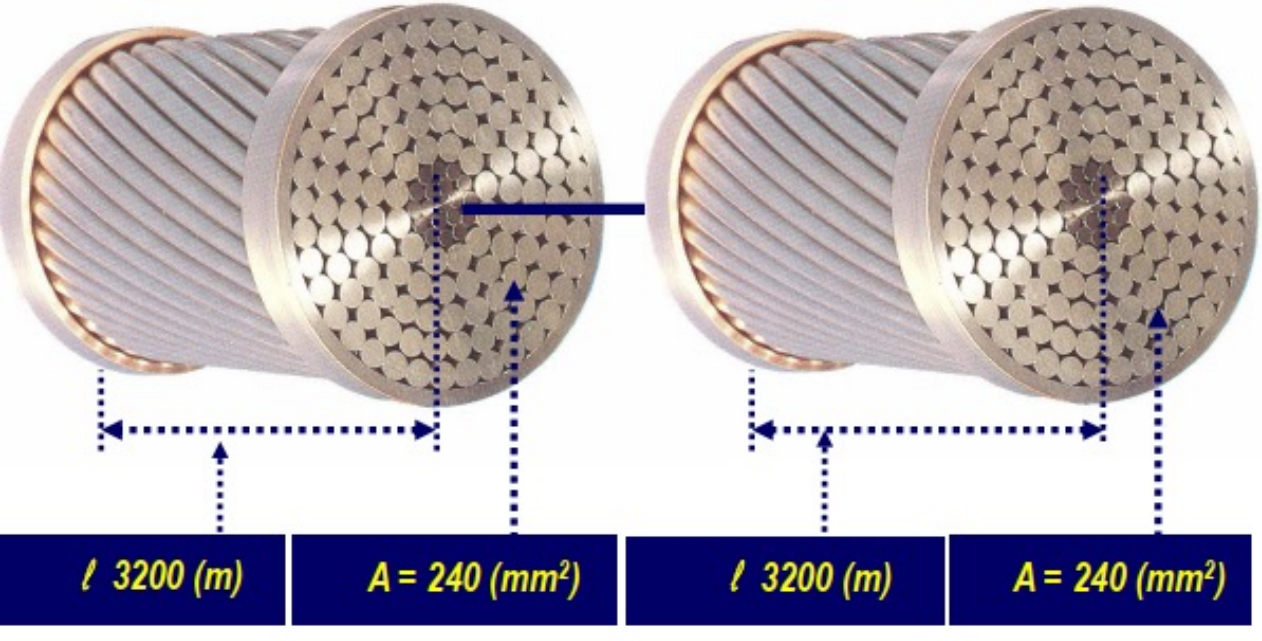
$$R_{total} = \rho \ell_{total} / A$$
$$= \rho (\ell_1 + \ell_2) / A$$
$$= \rho \ell_1 / A + \rho \ell_2 / A$$
$$= R_1 + R_2$$

Basic Principles

Series Connected Resistances

Equivalent Resistance Formula

$R_{total} = R_1 + R_2$	Series connected resistances are added





$$R_{total} = R_1 + R_2 + \ldots + R_k$$

Basic Principles

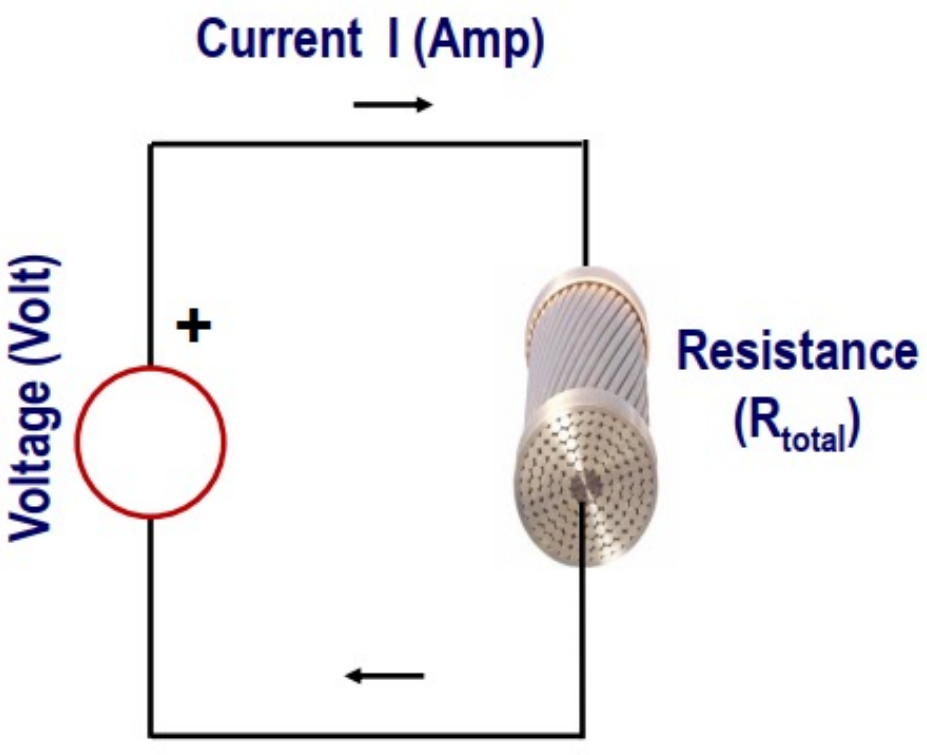
Ohm Law for Series Resistances

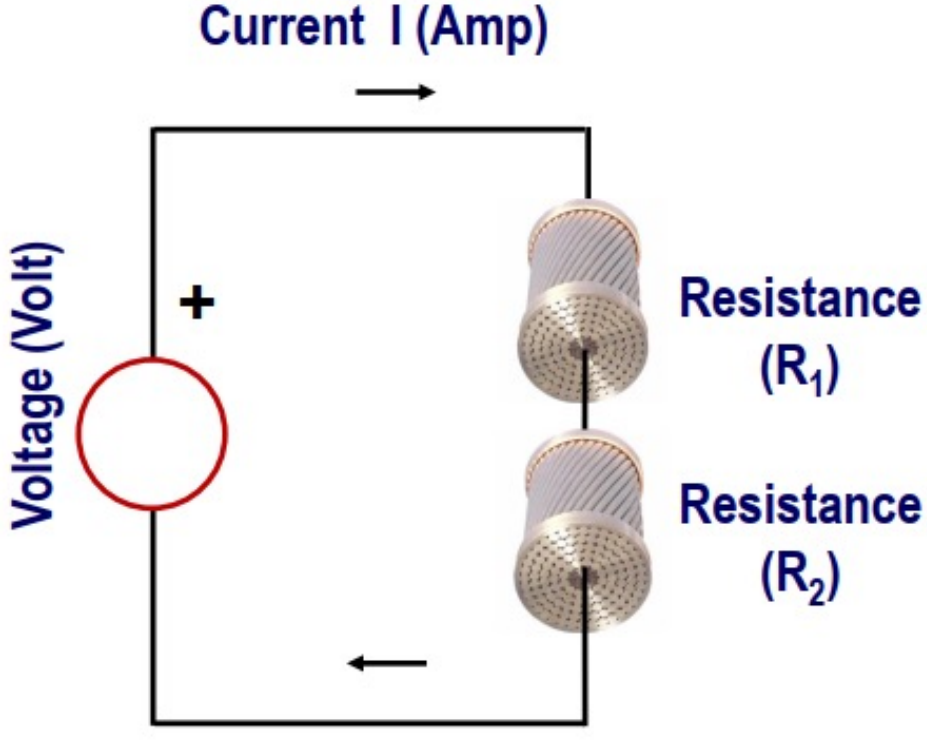
$V = R_{total} \times I$

(Volt) (Ohm) (Amp)

$V = R_1 \times I + R_2 \times I$

(Volt) (Volt) (Volt)

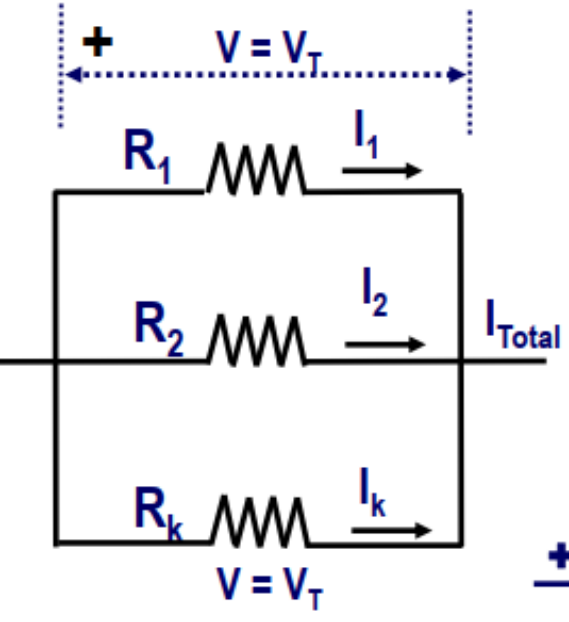




Basic Principles

Shunt Connected Resistances

Equivalent Resistance Formula



$$I_1 = V_T / R_1$$

$$I_2 = V_T / R_2$$

$$I_k = V_T / R_k$$

$$+$$

$$I_{total} = V_T(1 / R_1 + 1 / R_2 + \ldots + 1 / R_k) = V_T / R_{equivalent}$$

Hence,

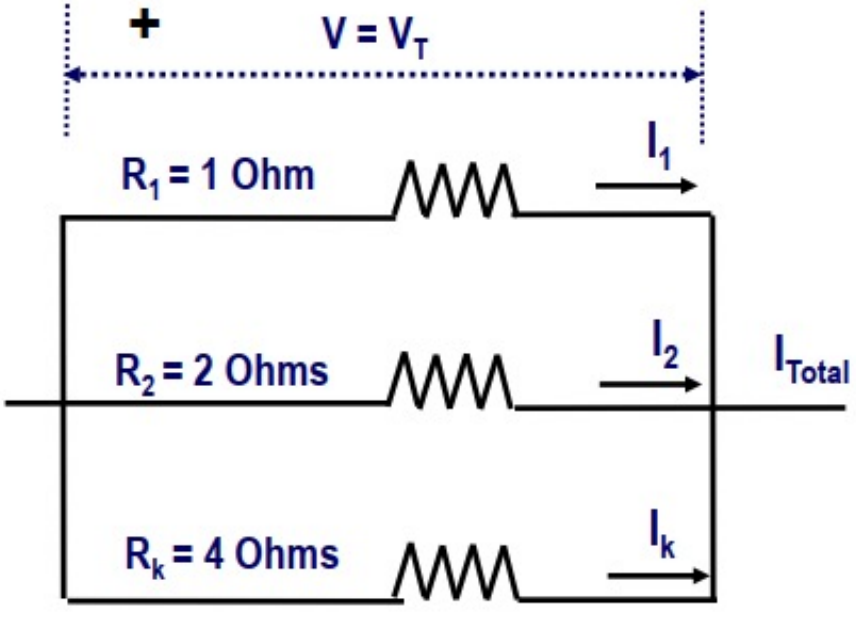
$$R_{equiv} = \frac{1}{1 / R_1 + 1 / R_2 + \ldots + 1 / R_k}$$

Basic Principles

Shunt Connected Resistances

Example

Find the equivalent resistance of the following connection



$$R_{equiv} = \frac{1}{1/R_1 + 1/R_2 + \ldots + 1/R_k}$$

$$R_{equiv} = \frac{1}{1/1 + 1/2 + 1/4}$$

$$= 1/(7/4) = 4/7 = \underline{0.5714 \text{ Ohm}}$$

Basic Principles

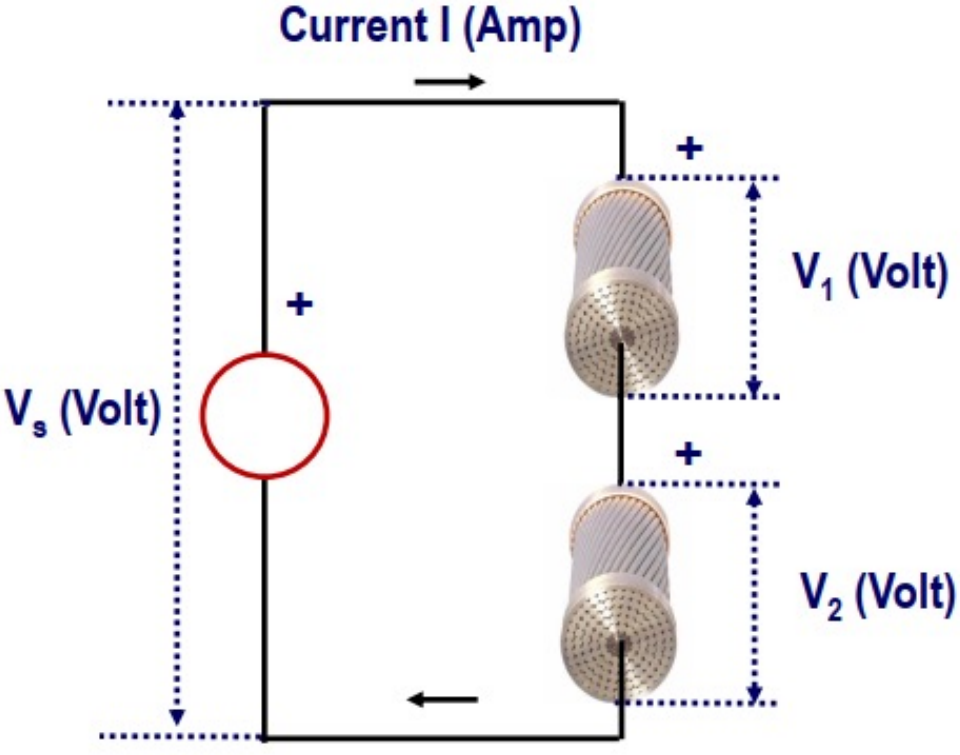
Voltages on Series Connected Elements

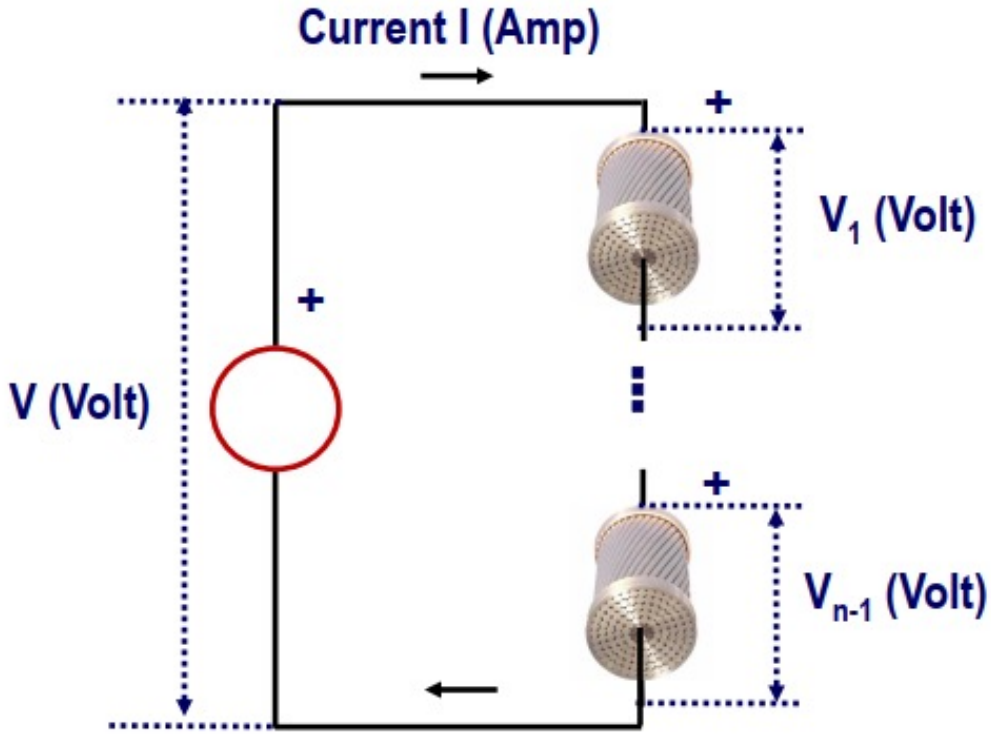
Voltages on series connected elements are added

$$V_{(Volt)} = V_{1\,(Volt)} + V_{2\,(Volt)}$$

or generalizing

$$V_{(Volt)} = V_{1\,(Volt)} + \ldots + V_{n-1\,(Volt)}$$





Basic Principles

Voltages on Series Connected Elements

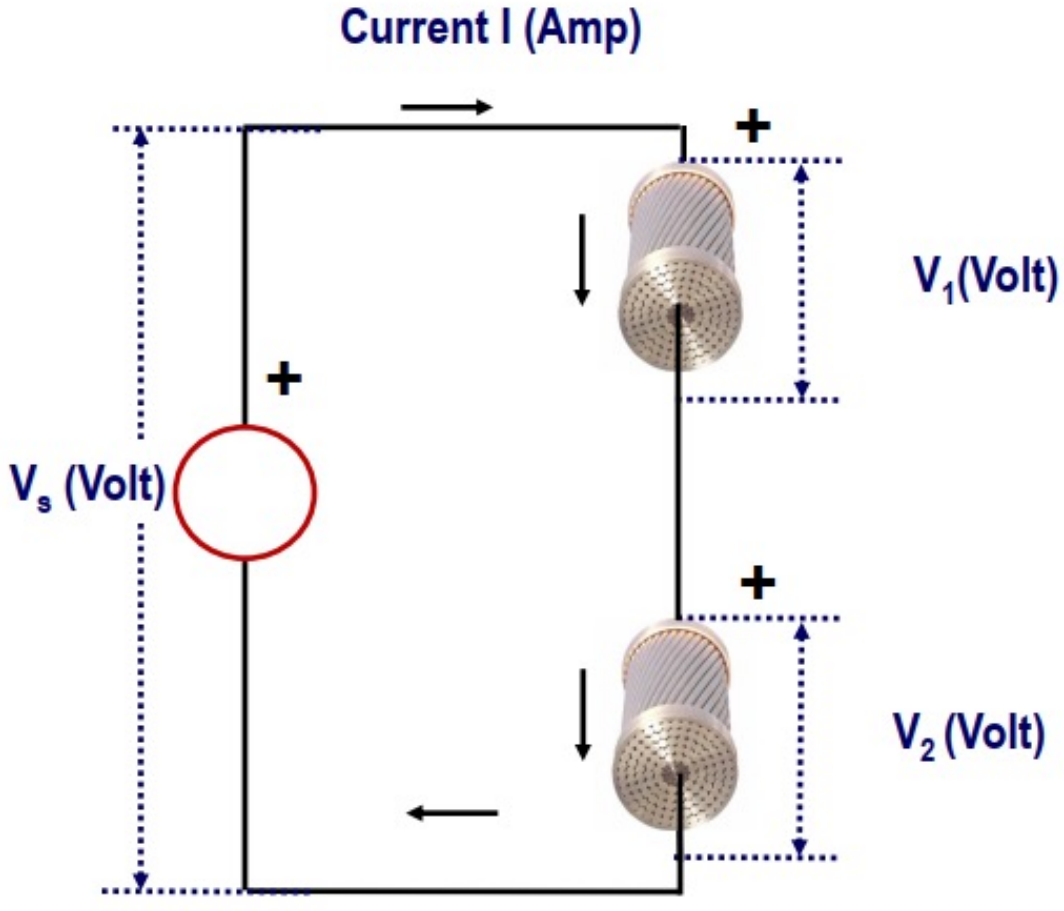
Voltages on series connected elements are added

$$V \;=\; V_1 + \ldots + V_{n-1}$$
(Volt) (Volt) (Volt)

$$V = \sum_{i=1}^{i=n-1} V_i$$

$$V_n - \sum_{i=1}^{i=n-1} V_i = 0$$

$$\sum_{i=1}^{i=n} V_i = 0$$



Basic Principles

Kirchoff's Voltage Law (KVL)

Statement

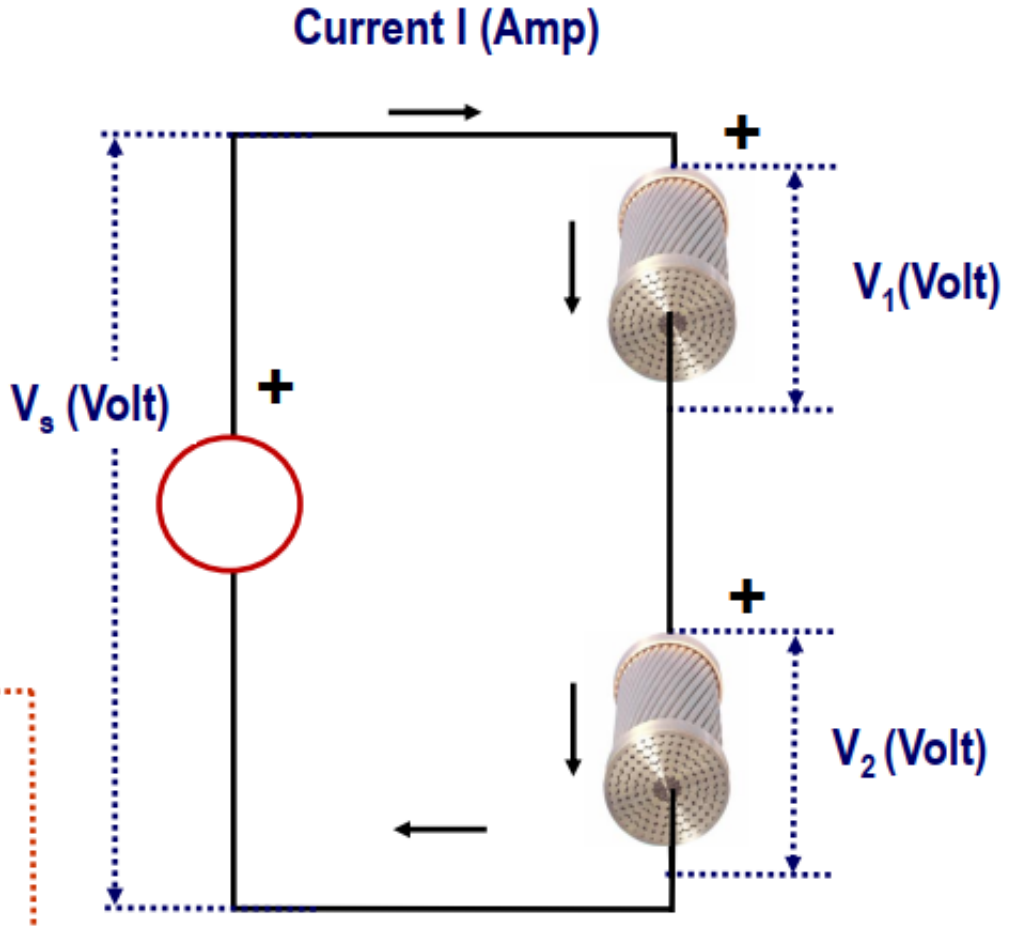
The above result may be expressed as;

Sum of voltages in a closed loop is zero

or

$$\sum_{i=1}^{i=n} V_i = 0$$

Kirchoff's Second Law
or
Kirchoff's Voltage Law



Basic Principles

Kirchoff's Voltage Law (KVL)

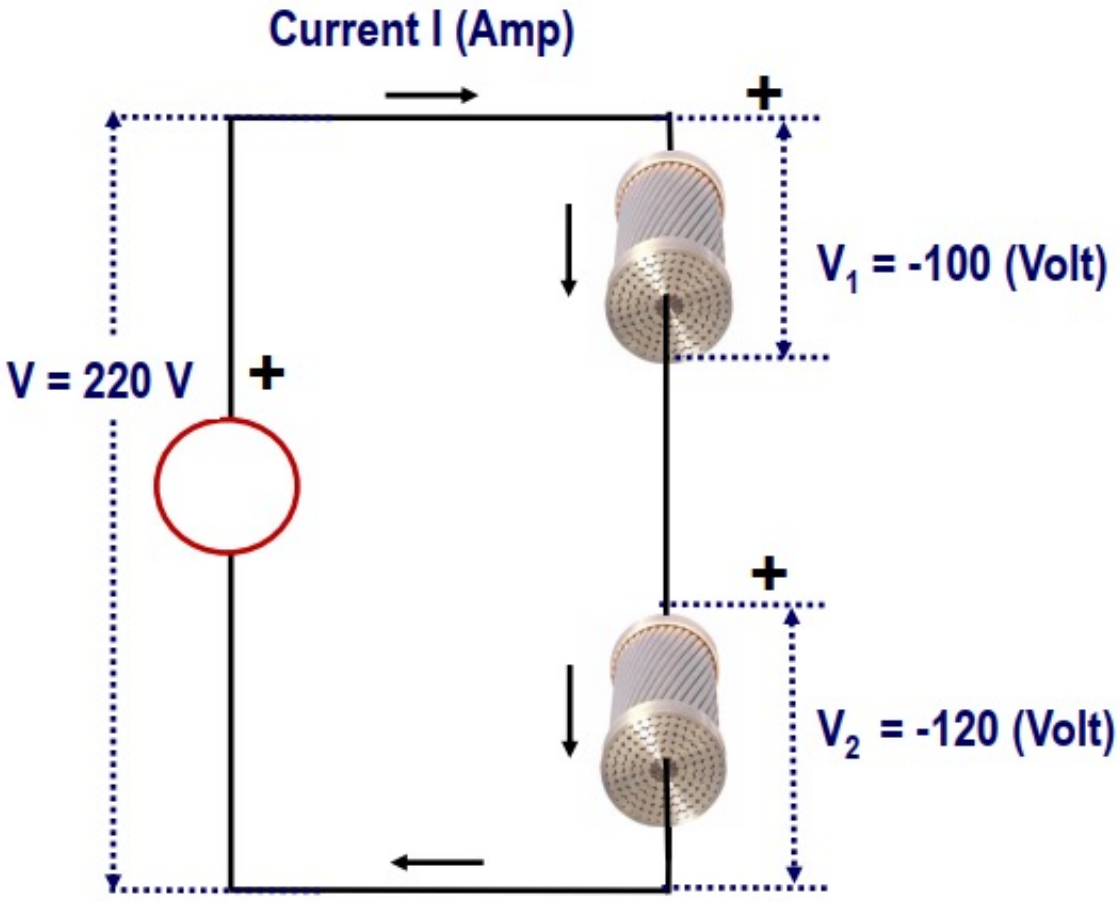
Example

$$\sum_{i=1}^{i=n} V_i = 0$$

$V_s = 220$ Volts

$V_s - V_1 - V_2 = 0$

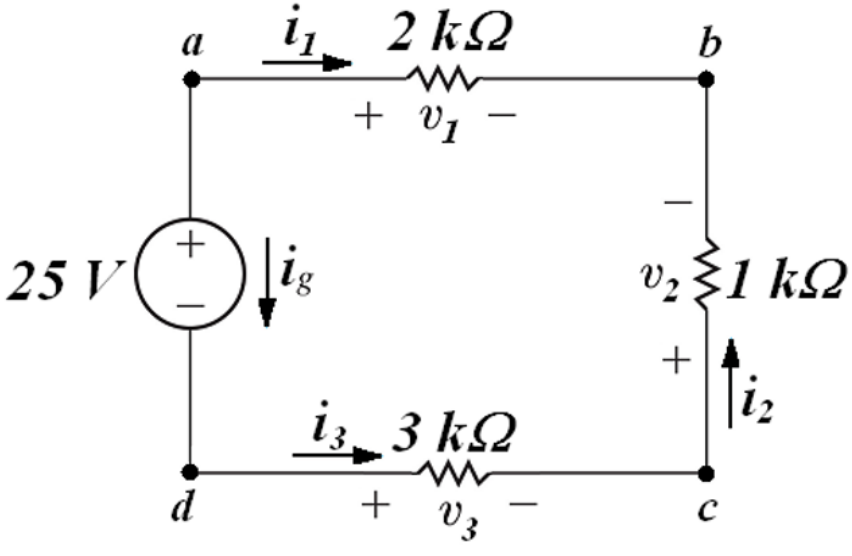
$220 - 100 - 120 = 0$



Basic Principles

Kirchhoff's Voltage Law (KVL)

- Let's write the KVL equation around the closed path formed by this circuit.



Start at node a and go clockwise:

$$+v_1 - v_2 - v_3 - 25 = 0$$

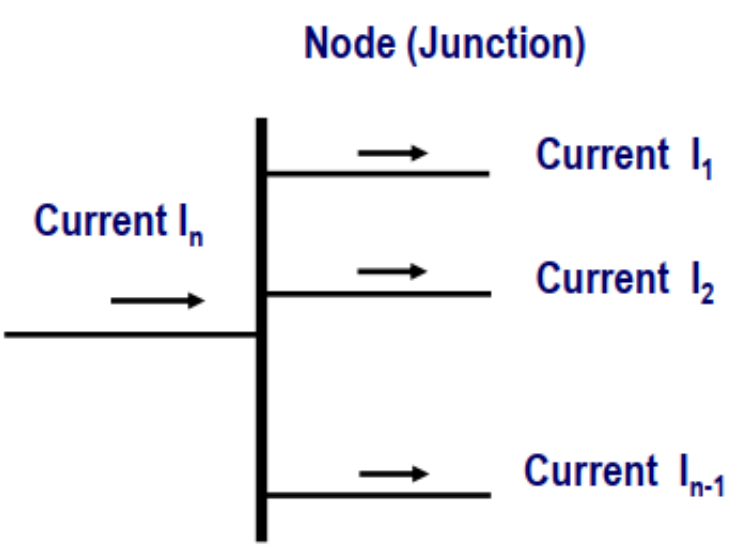
Start at node d and go counter-clockwise:

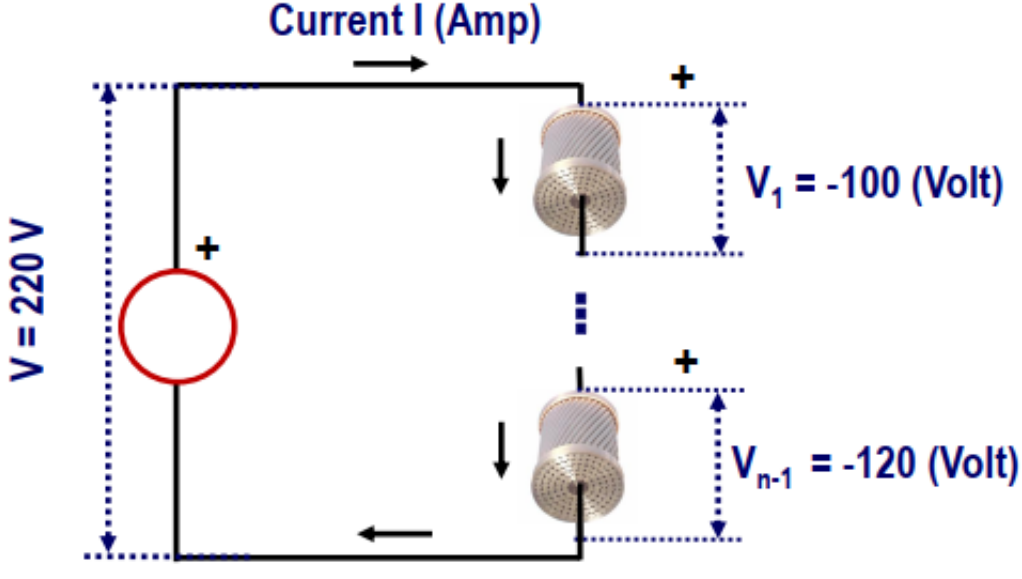
$$+v_3 + v_2 - v_1 + 25 = 0$$

Basic Principles

Summary of Kirchoff's Laws

Kirchoff's Current Law (KCL)	Kirchoff's Voltage Law (KVL)
Algebraic sum of currents entering a junction is zero	*Algebraic sum of voltages in a closed loop is zero*
$\sum_{i=1}^{i=n} I_i = 0$	$\sum_{i=1}^{i=n} V_i = 0$

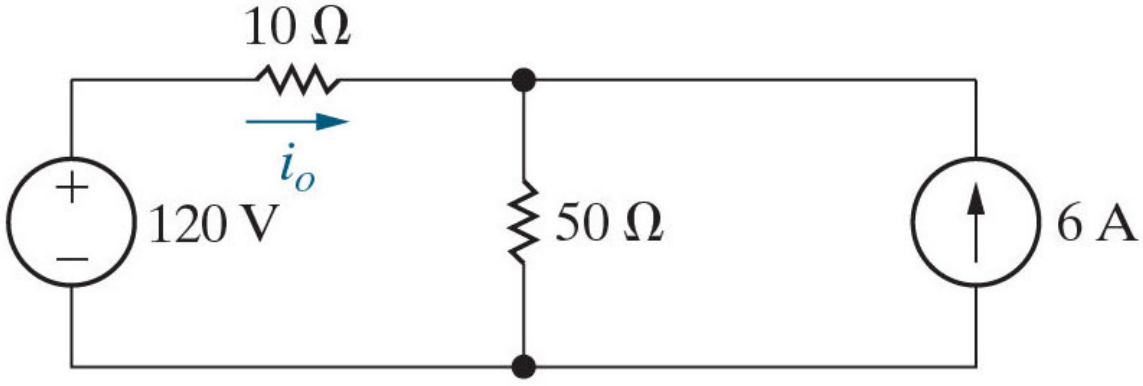




Circuit Elements

Example:

Find the current i_o



Begin by labeling all of the unknown voltages and currents and all of the nodes on the copy of this circuit on the next slide. Figure out how many unknowns you have. Then use Ohm's Law, KCL, and KVL to write the independent equations you need.

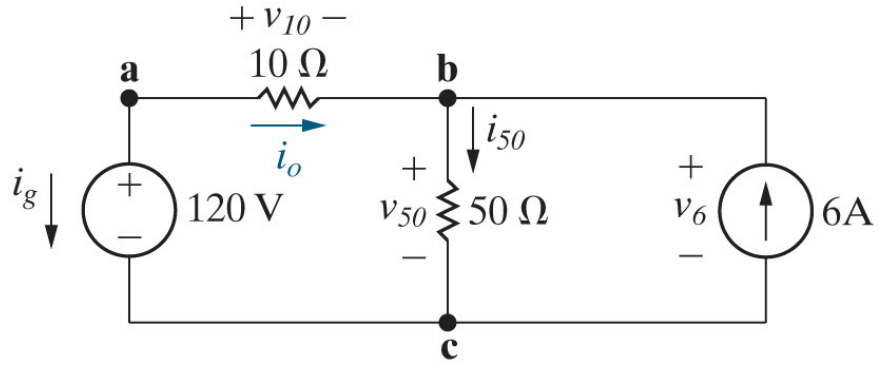
Chapter 2 – Circuit Elements (3 of 9)

Example (continued)

3 unknown voltages and

3 unknown currents

So we need 6 equations



Ohm's Law: $v_{50} = 50i_{50}$; $v_{10} = 10i_o$

KCL: At a: $i_g + i_o = 0$; At b: $i_o + 6 = i_{50}$

KVL: left: $-120 + v_{10} + v_{50} = 0$; right: $-v_{50} + v_6 = 0$

Solve KCL at b for i_{50}: $i_{50} = i_o + 6$

Substitute Ohm's Law equations in KVL left:

$$10i_o + 50i_{50} = 120$$

Substitute for i_{50} and solve:

$$10i_o + 50(i_o + 6) = 120 \quad \Rightarrow \quad 60i_o = -180$$

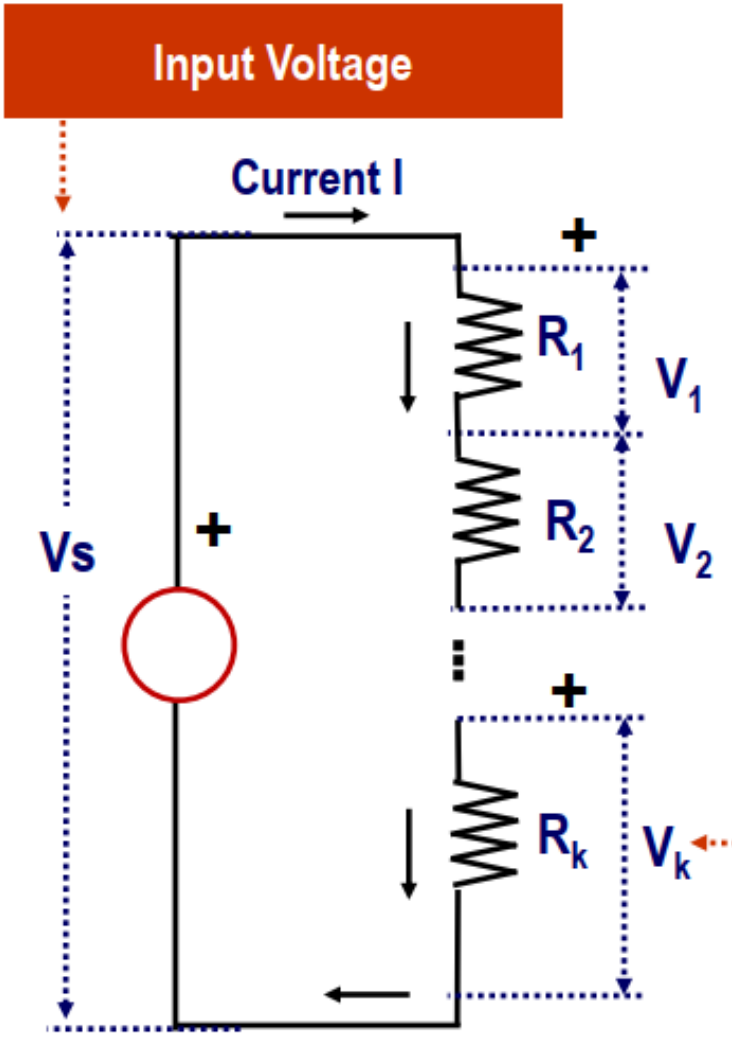
$$\Rightarrow \quad i_o = -3 \text{ A}$$

Basic Principles

Voltage Division Principle

$$V_1 = R_1 \times I$$
$$V_2 = R_2 \times I$$
$$\ldots$$
$$V_k = R_k \times I$$
$$+$$
$$V_s = V_1 + V_2 + \ldots V_k = (R_1 + \ldots + R_k) \times I$$
$$V_k / V_s = R_k / (R_1 + \ldots + R_k)$$

$$\text{Voltage Division Ratio} = \frac{R_k}{R_1 + \ldots + R_k}$$





End of Lecture

Next Week

We will continue with

- Node Analysis
- Mesh Analysis